

LAMP: Linear Verification of Matrix Multiplication via Proximity Testing

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Abstract—Verifiable computation systems often need to prove large matrix multiplication statements, but a direct SNARK arithmetization of a $k \times k$ product requires $\mathcal{O}(k^3)$ constraints. Freivalds’ randomized check reduces the algebraic computation to vector-matrix products, but proving those products inside a SNARK still costs $\mathcal{O}(k^2)$ constraints.

We present **LAMP**, a matrix-multiplication checking protocol that combines Freivalds’ randomized check with proximity testing over linear error-correcting codes. The prover commits to encoded matrices and intermediate vectors before the sampled query positions are derived. The CP-SNARK circuit then checks only the sampled codeword positions and commits to the values used inside the circuit, while Merkle openings and CP-Link proofs ensure consistency between the in-circuit witnesses and the externally committed values. We prove soundness for this committed-input setting under the soundness of the SNARK backend, the binding of the commitments, the correctness of the CP-Link checks, and the distance of the code.

For a fixed number t of sampled positions, the main in-circuit SNARK relation has $\mathcal{O}(tk)$ constraints, with additional $\mathcal{O}(t \log n) + E_{\text{link}}(k)$ backend work for Merkle openings and CP-Link checks. We implement **LAMP** in Go and compare it with a Freivalds-based SNARK circuit. In the matrix benchmark at $k = 2^{12}$, **LAMP** reduces the constraint count by $30.3\times$ and shortens proof generation time by $8.34\times$; verification stays below 0.01 seconds in the measured range.

Index Terms—Verifiable computation, SNARKs, Proximity testing, Linear codes, Freivalds’ algorithm, Merkle trees, Commitment linking

1. Introduction

Matrix multiplication is a core computational primitive underlying many large-scale computations, including numerical linear algebra, signal processing, computer graphics, and artificial intelligence. Consequently, many applications require verifiability for matrix products while keeping the underlying matrices private. Zero-knowledge proofs provide a cryptographic mechanism for this goal: a prover can convince a verifier that a statement is correct without revealing any secret information that witnesses the statement.

Applying general-purpose ZKP systems directly to large matrix multiplication, however, remains expensive. Succinct proof systems such as [1]–[4] prove relations after the computation has been represented as an arithmetic circuit or a similar algebraic constraint system. A standard

arithmetization of a $k \times k$ matrix product $\mathbf{AB} = \mathbf{C}$ computes every output entry and therefore uses $\mathcal{O}(k^3)$ arithmetic constraints. This is already impractical for moderate k , and it is especially problematic for Transformer-based neural networks [5]–[7], whose projection and attention layers often contain matrix multiplications with dimension $k \geq 1,000$.

The $\mathcal{O}(k^2)$ barrier. Freivalds’ classical randomized check [8] reduces matrix-product verification from a cubic computation to a quadratic one. Given matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}$ and a random vector $\mathbf{r}$, the verifier checks

$$\mathbf{rAB} = \mathbf{rC}$$

instead of recomputing the full product. This reduces the verification cost from $\mathcal{O}(k^3)$ to $\mathcal{O}(k^2)$. The resulting $\mathcal{O}(k^2)$ -size circuit is an improvement, but it is still expensive: for example, at $k = 2^{12}$, it already requires 50.5 million R1CS constraints and 405.04 s of proving time with Groth16. Thus, a large gap remains between the $\mathcal{O}(k^2)$ SNARK cost and practical large-scale matrix-multiplication verification.

Our insight: from computation to checking. We observe that the $\mathcal{O}(k^2)$ barrier is not inherent to the verification problem itself. It arises because existing SNARK-based approaches still *compute* the vector–matrix products inside the circuit. Our key idea is to move from “computing inside the circuit” to “*checking* inside the circuit.” The prover performs the expensive matrix arithmetic outside the SNARK and commits to the relevant encoded data and intermediate values. The verifier then checks only a small number of lightweight algebraic relations that are sufficient to detect inconsistent claims. The technical core of this approach is proximity testing over linear error-correcting codes. Let $\mathcal{C}$ be a linear code with relative distance δ . We encode each matrix row-wise and consider the Freivalds intermediate vectors

$$\mathbf{x} = \mathbf{rA}, \quad \mathbf{y} = \mathbf{xB}, \quad \mathbf{z} = \mathbf{rC}.$$

The linearity of $\mathcal{C}$ lets us check these vector–matrix products in the encoded domain, column by column. For one encoded column, a single inner product, or *fold*, checks one coordinate of the encoded product. If the prover cheats in one of the corresponding relations, the two relevant codewords disagree on at least a δ -fraction of positions. A random query therefore detects the inconsistency with probability at least δ .

The key point is that the SNARK circuit never takes the full matrices as witnesses. Instead, it checks fold equations

on t sampled encoded columns, where t is independent of the matrix dimension k . It also checks that the committed encoded views of the intermediate vectors $\mathbf{x}, \mathbf{y}, \mathbf{z}$ are valid codewords for those vectors. To bind the circuit witnesses to the data fixed before the query positions are known, we use a commit-and-prove SNARK for the sampled block witnesses. The prover also supplies Merkle membership proofs for the queried leaves under $\mathbf{rt}_{ABC}$ and $\mathbf{rt}_{XYZ}$, together with CP-Link proofs showing that the SNARK-side sampled blocks and the opened Pedersen leaves contain the same data. These checks ensure, under the backend assumptions, that the values checked inside the circuit are consistent with commitments fixed before the verifier’s sampling challenges.

As a result, the main SNARK relation has $\mathcal{O}(tk + E_{\text{code}})$ constraints, where E_{code} is the cost of checking that the intermediate vectors are valid encodings under the error-correcting code. For the fixed-rate Reed–Solomon setting used in our implementation, this remains linear in the matrix dimension for fixed t . The sampled opening and linking layer adds $\mathcal{O}(t \log n) + E_{\text{link}}(k, t)$ backend work. In our implementation, verification consists of SNARK verification, CP-Link verification, and $\mathcal{O}(t \log n)$ hash operations for Merkle openings; for the QALink instantiation, CP-Link verification includes $\mathcal{O}(t)$ pairing terms batched into a multi-pairing check.

Contributions. We present LAMP, a protocol for efficiently proving matrix-multiplication statements. Our contributions are as follows.

- **A code-based matrix-product checking protocol.** LAMP combines Freivalds’ randomized reduction with proximity testing over linear error-correcting codes. Instead of computing vector–matrix products inside the SNARK circuit, the protocol checks sampled fold equations over encoded commitments and verifies code consistency for the intermediate vectors. For fixed t , the main circuit cost is $\mathcal{O}(tk + E_{\text{code}})$, which is linear in k for the fixed-rate Reed–Solomon setting used in our implementation.
- **Soundness analysis.** We prove soundness for the committed-input setting. If $\mathbf{AB} \neq \mathbf{C}$, then any prover that commits to inconsistent encoded values is detected with probability governed by the code distance δ and the number t of sampled positions. The analysis also accounts for Merkle openings and CP-Link proofs, which bind the values checked inside the SNARK circuit to the data committed before the sampling challenges are derived.
- **Implementation and evaluation.** We implement LAMP in Go and compare it with a Freivalds-based SNARK circuit for matrix dimensions $k = 2^7$ through 2^{13} . At $k = 2^{12}$, LAMP reduces the constraint count by $30.3\times$ and proving time by $8.43\times$, while verification remains independent of the matrix dimension and stays around 0.06 s.

2. Related Work

Matrix multiplication.

Pinocchio [9] and Groth16 [1] provide constant-size proofs and constant verifier time, but directly applying them to matrix multiplication requires arithmetizing the entire computation. Thus, verifying the multiplication of two $k \times k$ matrices requires $\mathcal{O}(k^3)$ constraints, resulting in $\mathcal{O}(k^3 \log k)$ prover computation.

To overcome this limitation, several proof systems have studied how to verify matrix multiplication more efficiently without directly arithmetizing the full cubic computation. LegoSNARK [10] extends Groth16 with a commit-and-prove mechanism that ensures consistency between committed values and the SNARK witness. Combined with Freivalds’ randomized verification, matrix multiplication can be expressed using $\mathcal{O}(k^2)$ constraints, resulting in $\mathcal{O}(k^2 \log k)$ prover complexity while retaining constant-size proofs and constant verifier time. Thaler’s sum-check-based protocol [11] reduces matrix multiplication to checks over low-degree extensions, achieving $\mathcal{O}(k^2)$ prover work. However, the verifier computation still grows linearly with the matrix dimension, which can be costly for succinct verification of large matrices.

More recent works have developed specialized proof systems that verify matrix multiplication without relying on arithmetic circuits. zkMatrix [12] reduces matrix multiplication to four inner-product relations and adapts the techniques of Bulletproofs [13], achieving $\mathcal{O}(k^2)$ prover complexity and $\mathcal{O}(\log k)$ verifier complexity and proof size. DualMatrix [14], a follow-up work by the authors of zkMatrix, improves the prover complexity to $\mathcal{O}(K + k)$, where K denotes the number of non-zero entries. However, for dense matrices with $K = \Theta(k^2)$, its prover complexity remains $\mathcal{O}(k^2)$. zkMaP [15] employs KZG commitments to achieve $\mathcal{O}(k^2)$ prover work together with constant-size proofs and constant verifier time.

While these protocols achieve efficient verification of matrix multiplication, they are specialized to matrix multiplication itself. Therefore, in applications such as deep neural networks, where operations beyond matrix multiplication must be proved using a separate SNARK, an additional linking proof is required to ensure consistency between the witnesses used in the matrix multiplication proof and the SNARK.

Table 2 compares the computational complexities and proof characteristics of existing proof systems for matrix multiplication, including our approach.

Verifiable AI. Matrix multiplication is also a dominant cost in verifiable machine-learning systems. Systems such as vCNN [16], pvCNN [17], and zkVC [18] optimize verifiable inference for convolutional neural networks by reducing the number of multiplication constraints induced by QAP-based representations. Other systems use sum-check-style techniques. zkCNN [19] proposes a sum-check-based protocol for proving convolutions with linear prover time. In addition, zkGPT [20] and zkLLM [21]

System	Prover Work	Circuit Constraints	Verifier Work	Proof Size
Groth16	$\mathcal{O}(k^3 \log k)$	$\mathcal{O}(k^3)$	$\mathcal{O}(1)$	$\mathcal{O}(1)$
LegoSNARK	$\mathcal{O}(k^2 \log k)$	$\mathcal{O}(k^2)$	$\mathcal{O}(1)$	$\mathcal{O}(1)$
zkMatrix	$\mathcal{O}(k^2)$	–	$\mathcal{O}(\log k)$	$\mathcal{O}(\log k)$
DualMatrix	$\mathcal{O}(K + k)$	–	$\mathcal{O}(\log k)$	$\mathcal{O}(\log k)$
zkMaP	$\mathcal{O}(k^2)$	–	$\mathcal{O}(1)$	$\mathcal{O}(1)$
LAMP	$\mathcal{O}(k^2)$	$\mathcal{O}(tk)$	$\mathcal{O}(t \log k)$	$\mathcal{O}(t \log k)$

TABLE 1. ASYMPTOTIC COMPARISON WITH SYSTEMS FOR MATRIX MULTIPLICATION. HERE, $K = \Theta(k^2)$ FOR DENSE MATRICES AND t IS CONSTANT.

introduce protocols specialized for LLM architectures and use sum-check protocols to prove the correctness of linear layers efficiently. Meanwhile, Mystique [22] and zkML [23] use Freivalds’ randomized checks.

Code-based proximity testing. LAMP also builds on the use of linear error-correcting codes and proximity tests in succinct proof systems. FRI [24] and related works [25], [26] construct proximity IOPs for Reed–Solomon codewords. Recent work further extends this viewpoint to more general foldable codes [27]. Our proximity layer is closer in spirit to the local codeword tests used in Ligerio [28], Brakedown [29], and Blaze [30]. In particular, our use of in-circuit encoding checks is similar in spirit to Orion [31] and related work [32].

3. Preliminaries

We introduce notation and recall the building blocks used in our construction. Throughout, $\mathbb{F}$ denotes a finite field of size $|\mathbb{F}| = p$ for a prime p , and λ denotes the security parameter. We write $\text{negl}(\lambda)$ for any function that is negligible in λ , and PPT for probabilistic polynomial-time.

3.1. Notation

Matrices are denoted by bold uppercase letters $\mathbf{A}, \mathbf{B}, \mathbf{C} \in \mathbb{F}^{k \times k}$, and vectors by bold lowercase letters $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{F}^k$. For a vector $\mathbf{x}$, we write $\mathbf{x}[i]$ for its i -th entry. For a matrix $\mathbf{M}$, we write $\mathbf{M}[i, j]$ for the entry in the i -th row and j -th column. We denote the i -th row of $\mathbf{M}$ by $\mathbf{M}[i, *]$, and we will often write it simply as $\mathbf{M}[i]$. Similarly, we denote the j -th column of $\mathbf{M}$ by $\mathbf{M}[:, j]$. We write $[n] = \{0, 1, \dots, n-1\}$. For $t \in \mathbb{N}$, $I \leftarrow \$ [n]^t$ denotes the uniform sampling of t indices from $[n]$. Similarly, $r \leftarrow \$ \mathbb{F}$ denotes uniform random sampling from $\mathbb{F}$.

3.2. Linear Error-Correcting Codes

A *linear code* over $\mathbb{F}$ is an $\mathbb{F}$ -linear map $\mathcal{C} : \mathbb{F}^k \rightarrow \mathbb{F}^n$ defined by a generator matrix $\mathbf{G} \in \mathbb{F}^{k \times n}$, so that $\mathcal{C}(\mathbf{m}) = \mathbf{m} \cdot \mathbf{G}$ for any input vector $\mathbf{m} \in \mathbb{F}^k$. The code has *rate* $\rho = k/n$ and *minimum distance* $d = \min_{\mathbf{c} \neq \mathbf{c}'} \Delta(\mathbf{c}, \mathbf{c}')$, where the minimum is taken over distinct codewords $\mathbf{c}, \mathbf{c}' \in \mathcal{C}$, and Δ denotes the Hamming distance.

The *relative distance* is $\delta = d/n$. The unique-decoding radius is $\tau_{\text{UD}} = \frac{1}{n} \lfloor \frac{d-1}{2} \rfloor$, within which a received word has at most one nearby codeword. We use a proximity threshold $\tau \leq \tau_{\text{UD}}$ throughout the protocol.

We write $\text{Enc}(\mathbf{m}) := \mathcal{C}(\mathbf{m})$ for the encoding of an input vector. For a matrix $\mathbf{M} \in \mathbb{F}^{k \times k}$, we define the *row-wise encoding* $\widehat{\mathbf{M}} := \text{Enc}(\mathbf{M}) \in \mathbb{F}^{k \times n}$ as the matrix whose i -th row is $\text{Enc}(\mathbf{M}[i, *])$ for each $i \in [k]$. The j -th column of $\widehat{\mathbf{M}}$ is denoted by $\widehat{\mathbf{M}}[:, j] \in \mathbb{F}^k$ for each $j \in [n]$.

Definition 3.1 (Interleaved code). For an integer $\ell \geq 1$, the ℓ -fold interleaved code $\mathcal{C}^{(\ell)}$ is obtained by grouping ℓ codewords of $\mathcal{C}$ position by position. More concretely, for codewords

$$\mathbf{c}^{(0)}, \dots, \mathbf{c}^{(\ell-1)} \in \mathcal{C},$$

their interleaving is the word $\mathbf{w} \in (\mathbb{F}^\ell)^n$ whose j -th symbol is

$$\mathbf{w}[j] = (\mathbf{c}^{(0)}[j], \dots, \mathbf{c}^{(\ell-1)}[j]) \in \mathbb{F}^\ell \quad \text{for } j \in [n].$$

The code $\mathcal{C}^{(\ell)}$ is the set of all such interleavings.

The row-wise encoding of a matrix is an interleaved codeword in this sense. Indeed, for $\mathbf{M} \in \mathbb{F}^{k \times k}$, the rows $\text{Enc}(\mathbf{M}[0, *]), \dots, \text{Enc}(\mathbf{M}[k-1, *])$ are k codewords of $\mathcal{C}$, and their interleaving is exactly $\widehat{\mathbf{M}} \in \mathbb{F}^{k \times n}$. The j -th interleaved symbol is the encoded column $\widehat{\mathbf{M}}[:, j] \in \mathbb{F}^k$.

The key property we exploit is *linearity*: for any coefficients $r_0, \dots, r_{k-1} \in \mathbb{F}$,

$$\sum_{i=0}^{k-1} r_i \cdot \widehat{\mathbf{M}}[i, *] = \text{Enc} \left(\sum_{i=0}^{k-1} r_i \cdot \mathbf{M}[i, *] \right) \quad (1)$$

Definition 3.2 (Fold). For vectors $\mathbf{c}, \mathbf{r} \in \mathbb{F}^k$ with $\mathbf{c} = (c_0, \dots, c_{k-1})^\top$ and $\mathbf{r} = (r_0, \dots, r_{k-1})$, define

$$\text{Fold}(\mathbf{r}, \mathbf{c}) := \langle \mathbf{r}, \mathbf{c} \rangle = \sum_{i=0}^{k-1} r_i \cdot c_i.$$

Equivalently, for the j -th column of a row-wise encoded matrix, $\text{Fold}(\mathbf{r}, \widehat{\mathbf{M}}[:, j])$ equals the j -th coordinate of $\text{Enc}(\mathbf{rM})$. This follows directly from the linearity of the code (Equation 1).

Concrete instantiations. Our protocol is compatible with any linear code. Two natural choices are: (i) *Reed–Solomon codes*, where $\mathcal{C}(\mathbf{m})$ evaluates the polynomial $\sum_i m_i X^i$ over a domain of size $n > k$, achieving the optimal (MDS) distance $\delta = 1 - k/n + 1/n$; and (ii) *expander-based codes* as used in Brakedown [33], which achieve linear-time encoding $\mathcal{O}(n)$ with constant relative distance.

3.3. Commitment Schemes

A commitment scheme allows a committer to bind itself to a message and later open it. Some commitment schemes also hide the message before opening; our soundness arguments use only binding. In this work, we use two types of commitments: Merkle tree commitments and Pedersen vector commitments.

3.3.1. Merkle tree commitments. A Merkle tree commitment over a collision-resistant hash function $H : \{0, 1\}^* \rightarrow \{0, 1\}^\lambda$ is a vector commitment for a sequence $\mathbf{v} = (\mathbf{v}[j])_{j \in [n]}$. The leaves may be randomized or deterministic. We write $\text{MT.Com}(\mathbf{v}; \mathbf{o})$ for randomized leaves and $\text{MT.Com}(\mathbf{v}; \perp)$ for deterministic leaves.

- $\text{rt} \leftarrow \text{MT.Com}(\mathbf{v}; \mathbf{o})$: outputs a root hash rt obtained by constructing a binary hash tree with randomized leaves

$$\ell_j = \begin{cases} H(\mathbf{v}[j], \mathbf{o}[j]) & \text{if } \mathbf{o} \neq \perp, \\ H(\mathbf{v}[j]) & \text{if } \mathbf{o} = \perp. \end{cases}$$

Internal nodes are computed by hashing their two children.

- $(\mathbf{v}[j], o_j, \pi_j) \leftarrow \text{MT.Open}(\mathbf{v}, \mathbf{o}, j)$: outputs the leaf value $\mathbf{v}[j]$, its opening randomness o_j , and its authentication path π_j for position $j \in [n]$. In the deterministic case $\mathbf{o} = \perp$, we omit the randomness and write $(\mathbf{v}[j], \pi_j) \leftarrow \text{MT.Open}(\mathbf{v}, j)$.
- $b \leftarrow \text{MT.Verify}(\text{rt}, j, \mathbf{v}[j], \mathbf{o}[j], \pi_j)$: recomputes the randomized leaf $\ell_j = H(\mathbf{v}[j], \mathbf{o}[j])$ and outputs $b = 1$ if the authentication path is consistent with root rt at position j , and outputs $b = 0$ otherwise. In the deterministic case, we write $\text{MT.Verify}(\text{rt}, j, \mathbf{v}[j], \pi_j)$.

Merkle tree commitments are *position-binding*: under the collision resistance of H , no PPT adversary can produce a root rt , an index $j \in [n]$, two distinct openings $(v, o) \neq (v', o')$, and authentication paths π_j, π'_j such that

$$\text{MT.Verify}(\text{rt}, j, v, o, \pi_j) = \text{MT.Verify}(\text{rt}, j, v', o', \pi'_j) = 1,$$

except with probability $\epsilon_{\text{merkle}}(\lambda) = \text{negl}(\lambda)$.

3.3.2. Pedersen commitments. Pedersen commitments can be viewed as algebraic vector commitments over a group $\mathbb{G}$ of prime order q . Let $\text{ck} = (G_1, \dots, G_n, H)$ be a commitment key consisting of fixed generators in $\mathbb{G}$. A Pedersen commitment is defined by the following algorithms.

- $\text{cm} \leftarrow \text{Ped.Com}(\text{ck}, \mathbf{v}; o)$: outputs a commitment $\text{cm} := \sum_{i=1}^n \mathbf{v}[i]G_i + oH \in \mathbb{G}$ for a vector $\mathbf{v} \in \mathbb{F}^n$ and randomness $o \leftarrow \mathbb{F}$.

- $\pi \leftarrow \text{Ped.Open}(\mathbf{v}, o)$: outputs the opening $\pi = (\mathbf{v}, o)$ consisting of the committed vector and the randomness.
- $b \leftarrow \text{Ped.Verify}(\text{ck}, \text{cm}, \mathbf{v}, o)$: outputs $b = 1$ if $\text{cm} \stackrel{?}{=} \sum_{i=1}^n \mathbf{v}[i]G_i + oH$, and outputs $b = 0$ otherwise.

Pedersen commitments are *perfectly hiding* and *computationally binding*: under the discrete logarithm assumption, no PPT adversary can produce a commitment cm , two distinct vectors $\mathbf{v} \neq \mathbf{v}'$, and randomness values o, o' such that

$$\text{Ped.Verify}(\text{ck}, \text{cm}, \mathbf{v}, o) = \text{Ped.Verify}(\text{ck}, \text{cm}, \mathbf{v}', o') = 1,$$

except with probability $\epsilon_{\text{ped}}(\lambda) = \text{negl}(\lambda)$.

They are also *additively homomorphic*: for $\text{cm}_1 = \text{Ped.Com}(\text{ck}, \mathbf{v}_1; o_1)$ and $\text{cm}_2 = \text{Ped.Com}(\text{ck}, \mathbf{v}_2; o_2)$, $\text{cm}_1 + \text{cm}_2 = \text{Ped.Com}(\text{ck}, \mathbf{v}_1 + \mathbf{v}_2; o_1 + o_2)$.

3.4. Succinct Non-interactive Arguments of Knowledge

A SNARK (succinct non-interactive argument of knowledge) for an NP relation $\mathcal{R}$ is a tuple of algorithms $\Pi = (\text{Setup}, \text{Prove}, \text{Verify})$.

- $\text{pp} \leftarrow \Pi.\text{Setup}(1^\lambda, \mathcal{R})$: generates public parameters pp for the relation $\mathcal{R}$.
- $\pi \leftarrow \Pi.\text{Prove}(\text{pp}, \mathbf{x}, \mathbf{w})$: outputs a proof π for a statement $\mathbf{x}$ and witness $\mathbf{w}$ satisfying $(\mathbf{x}, \mathbf{w}) \in \mathcal{R}$.
- $b \leftarrow \Pi.\text{Verify}(\text{pp}, \mathbf{x}, \pi)$: outputs $b = 1$ if π is accepted as a valid proof for $\mathbf{x}$, and outputs $b = 0$ otherwise.

Remark 1 (Security Properties). We use the standard SNARK notions of completeness, knowledge soundness, and zero knowledge. Appendix A recalls these notions formally, including their negligible error bounds.

3.4.1. Commit-and-Prove SNARKs. A commit-and-prove SNARK (CP-SNARK) [10], [34] is a variant of a SNARK that proves statements about witnesses bound to an external commitment. We call such a commitment a *witness commitment*. Let Com be a commitment scheme with commitment key ck . For an NP relation $\mathcal{R}$ over tuples $(\mathbf{x}, \mathbf{u}, \omega)$, a CP-SNARK proves knowledge of a witness $\mathbf{w} := (\mathbf{u}, \omega)$ such that $(\mathbf{x}, \mathbf{u}, \omega) \in \mathcal{R}$, where $\mathbf{u}$ is the committed part of the witness and ω is the remaining private witness. The committed witness is bound separately by a witness commitment $\widetilde{\text{cm}} = \text{Com}(\text{ck}_S, m_0 || \dots || m_{q-1}; \tilde{o})$, where $\tilde{o}$ is the opening randomness. Equivalently, the CP proof is verified with respect to the extended public statement $\mathbf{x}_{\text{cp}} := (\mathbf{x}, \widetilde{\text{cm}})$. When the context is clear, we write this extended statement simply as $\mathbf{x}$. A CP-SNARK scheme is a tuple of algorithms $\Pi_{\text{cp}} = (\text{Setup}, \text{Com}, \text{Prove}, \text{Verify})$.

- $\text{pp} \leftarrow \Pi_{\text{cp}}.\text{Setup}(1^\lambda, \mathcal{R})$: generates public parameters $\text{pp} = (\text{pk}, \text{vk}, \text{ck})$ for the relation $\mathcal{R}$, where pk is the proving key, vk is the verifying key, and ck is the commitment key.
- $\widetilde{\text{cm}} \leftarrow \Pi_{\text{cp}}.\text{Com}(\text{pp}, \mathbf{u}; \tilde{o})$: outputs a witness commitment to the committed witness part $\mathbf{u}$.
- $\pi_{\text{cp}} \leftarrow \Pi_{\text{cp}}.\text{Prove}(\text{pp}, \mathbf{x}_{\text{cp}}, \mathbf{w})$: outputs a proof for the extended statement $\mathbf{x}_{\text{cp}}$ and witness $\mathbf{w} = (\mathbf{u}, \omega)$.

- $b \leftarrow \Pi_{\text{cp}}.\text{Verify}(\text{pp}, x_{\text{cp}}, \pi_{\text{cp}})$: outputs $b = 1$ if π_{cp} is accepted as a valid proof for the extended statement x_{cp} , and outputs $b = 0$ otherwise. Equivalently, we may write verification as $\Pi_{\text{cp}}.\text{Verify}(\text{pp}, x, \widetilde{\text{cm}}, \pi_{\text{cp}})$.

3.5. CP-Link

A CP-Link proof is a proof system for linking commitments. In the form used by LAMP, it proves that one SNARK-side commitment to an ordered concatenation of sampled blocks is consistent with external commitments to the individual blocks, without revealing the blocks or the opening randomness.

Let Com be a commitment scheme. For a SNARK-side key ck_S , an external key ck_E , a SNARK-side commitment $\widetilde{\text{cm}}$, and external commitments $\text{cm}_0, \dots, \text{cm}_{q-1}$, the blockwise linking relation is defined over the statement and witness

$$\begin{aligned} x_{\text{link}} &= ((\text{ck}_S, \widetilde{\text{cm}}), (\text{ck}_E, \text{cm}_0, \dots, \text{cm}_{q-1})), \\ w_{\text{link}} &= ((\mathbf{m}_i)_{i=0}^{q-1}, \widetilde{o}, (o_i)_{i=0}^{q-1}). \end{aligned}$$

The pair $(x_{\text{link}}, w_{\text{link}})$ is in $\mathcal{R}_{\text{link}}$ if and only if

$$\begin{aligned} \widetilde{\text{cm}} &= \text{Com}(\text{ck}_S, \mathbf{m}_0 \parallel \dots \parallel \mathbf{m}_{q-1}; \widetilde{o}), \\ \text{cm}_i &= \text{Com}(\text{ck}_E, \mathbf{m}_i; o_i) \quad \text{for all } i = 0, \dots, q-1. \end{aligned}$$

A proof for $\mathcal{R}_{\text{link}}$ convinces the verifier that the ordered blocks committed inside the CP-SNARK are exactly the blocks committed by the corresponding external commitments.

We use CP-Link through the standard statement/witness interface for a linking relation.

A CP-Link proof system is a tuple of algorithms $\Pi_{\text{link}} = (\text{Setup}, \text{Prove}, \text{Verify})$.

- $\text{pp}_{\text{link}} \leftarrow \Pi_{\text{link}}.\text{Setup}(1^\lambda, \mathcal{R}_{\text{link}})$: generates public parameters for the linking relation.
- $\pi_{\text{link}} \leftarrow \Pi_{\text{link}}.\text{Prove}(\text{pp}_{\text{link}}, x_{\text{link}}, w_{\text{link}})$: outputs a proof that the statement-witness pair satisfies the linking relation.
- $b \leftarrow \Pi_{\text{link}}.\text{Verify}(\text{pp}_{\text{link}}, x_{\text{link}}, \pi_{\text{link}})$: outputs $b = 1$ if the proof is accepted, and outputs $b = 0$ otherwise.

Remark 2 (Security Properties). For the soundness of LAMP, we require CP-Link soundness: an accepting linking proof should imply membership in $\mathcal{R}_{\text{link}}$, except with error $\epsilon_{\text{link}}(\lambda)$. If the linking backend is used to protect the linked blocks and openings, one may additionally require witness zero knowledge. Appendix B states these properties.

3.6. Freivalds' Algorithm

Freivalds' algorithm [8] is a classical randomized algorithm for verifying matrix multiplication. Given $\mathbf{A}, \mathbf{B}, \mathbf{C} \in \mathbb{F}^{k \times k}$, it samples $\mathbf{r} \leftarrow \mathbb{F}^k$ and checks whether $\mathbf{r}(\mathbf{AB}) = \mathbf{rC}$. The correctness relies on the following:

Lemma 3.3 (Schwartz–Zippel [35], [36]). *Let $P(X_1, \dots, X_m)$ be a non-zero polynomial of total*

degree d over a finite field $\mathbb{F}$. Then for $r_1, \dots, r_m \leftarrow \mathbb{F}$ chosen uniformly and independently,

$$\Pr[P(r_1, \dots, r_m) = 0] \leq \frac{d}{|\mathbb{F}|}.$$

In our protocol, we use a structured variant: instead of sampling $\mathbf{r} \leftarrow \mathbb{F}^k$ uniformly, we sample one scalar $r \leftarrow \mathbb{F}$ and set $\mathbf{r} = (1, r, r^2, \dots, r^{k-1})$. This keeps the Freivalds challenge compact in the transcript and in the SNARK statement: the verifier records one field element rather than k independent ones. If $\mathbf{D} = \mathbf{AB} - \mathbf{C} \neq \mathbf{0}$, then for any non-zero column $\mathbf{d}_j$ of $\mathbf{D}$, the value $\langle \mathbf{r}, \mathbf{d}_j \rangle$ is a non-zero univariate polynomial in r of degree at most $k-1$. By Lemma 3.3, it vanishes with probability at most $(k-1)/|\mathbb{F}|$.

4. Overview of LAMP

This section gives a technical overview of LAMP before the formal construction in Section 5. The main message is that LAMP does not ask the SNARK circuit to compute a matrix product. Instead, the prover commits to encoded data, and the circuit checks only a small number of local consistency equations at randomly sampled codeword positions.

4.1. Problem Statement and Relation Reduction Chain

The starting point is the matrix multiplication relation

$$\mathbf{AB} = \mathbf{C}, \quad \mathbf{A}, \mathbf{B}, \mathbf{C} \in \mathbb{F}^{k \times k}.$$

A direct SNARK arithmetization of this relation computes every entry of $\mathbf{AB}$. This requires k^3 scalar multiplications and therefore $\mathcal{O}(k^3)$ arithmetic constraints. This is the original relation:

$$\mathcal{R}_{\text{orig}} = \{(\mathbf{A}, \mathbf{B}, \mathbf{C}) : \mathbf{AB} = \mathbf{C}\}.$$

Freivalds' algorithm reduces the cubic check to a vector-matrix check. The classical algorithm samples a challenge vector $\mathbf{r} \in \mathbb{F}^k$ and checks $\mathbf{rAB} = \mathbf{rC}$.

In LAMP, the verifier derives a scalar $r \in \mathbb{F}$ and sets $\mathbf{r} = (1, r, \dots, r^{k-1})$, which keeps the public challenge compact. At the same transcript point, the verifier derives an independent scalar $s \in \mathbb{F}$ and sets $\mathbf{s} = (1, s, \dots, s^{k-1})$. The role of this additional challenge in testing the proximity of the committed encoding of $\mathbf{B}$ is described in Section 4.2

Equivalently, the prover introduces intermediate vectors

$$\mathbf{x} = \mathbf{rA}, \quad \mathbf{y} = \mathbf{xB}, \quad \mathbf{z} = \mathbf{rC},$$

and the verifier checks $\mathbf{y} = \mathbf{z}$. This gives the Freivalds relation

$$\begin{aligned} \mathcal{R}_{\text{Fr}} = \{ & (\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{x}, \mathbf{y}, \mathbf{z}) : \\ & \mathbf{x} = \mathbf{rA}, \quad \mathbf{y} = \mathbf{xB}, \quad \mathbf{z} = \mathbf{rC}, \quad \mathbf{y} = \mathbf{z} \}. \end{aligned}$$

This removes the cubic matrix-matrix computation, but a SNARK circuit that computes $\mathbf{rA}$, $\mathbf{xB}$, and $\mathbf{rC}$ directly still performs $\mathcal{O}(k^2)$ scalar multiplications.

LAMP applies one more reduction. Rather than computing the full vector-matrix products inside the circuit, it checks the products only at randomly sampled positions in an encoded domain. The prover performs the expensive native computation outside the SNARK, commits to the encoded matrices and intermediate vectors, and the SNARK checks local equations of the form

$$\text{Enc}(\mathbf{x})[j] = \text{Fold}(\mathbf{r}, \widehat{\mathbf{A}}[*], j),$$

$$\text{Enc}(\mathbf{y})[j] = \text{Fold}(\mathbf{x}, \widehat{\mathbf{B}}[*], j),$$

$$\text{Enc}(\mathbf{z})[j] = \text{Fold}(\mathbf{r}, \widehat{\mathbf{C}}[*], j).$$

The prover additionally computes $\mathbf{b} = \mathbf{sB}$, the circuit checks

$$\text{Enc}(\mathbf{b})[j] = \text{Fold}(\mathbf{s}, \widehat{\mathbf{B}}[*], j).$$

An additional fold based on the independent challenge $\mathbf{s}$ is used to test the proximity of the committed encoding of $\mathbf{B}$, as described in Section 4.2.

Only the queried columns $\widehat{\mathbf{A}}[*], j$, $\widehat{\mathbf{B}}[*], j$, $\widehat{\mathbf{C}}[*], j$ are used by the circuit. Thus, for t queried positions, the fold part of the circuit has cost $\mathcal{O}(tk)$, which is linear in k for fixed t .

4.2. Encoding and Proximity Testing

We use the proximity threshold $\tau \leq \tau_{\text{UD}}$ defined in Section 3.2. For a committed row-wise encoding, proximity is measured with respect to the corresponding interleaved code $\mathcal{C}^{(k)}$.

For a matrix $\mathbf{M} \in \mathbb{F}^{k \times k}$, the prover row-wise encodes $\mathbf{M}$ as $\widehat{\mathbf{M}} = \text{Enc}(\mathbf{M}) \in \mathbb{F}^{k \times n}$. By Definition 3.1, this row-wise encoding can also be viewed as a k -fold interleaved codeword, where the j -th interleaved symbol is the encoded column $\widehat{\mathbf{M}}[*], j$.

The key property underlying the local checks is the linearity of the code. For any $\mathbf{u} = (u_0, \dots, u_{k-1}) \in \mathbb{F}^k$ and every $j \in [n]$,

$$\text{Enc}(\mathbf{uM})[j] = \sum_{i=0}^{k-1} u_i \widehat{\mathbf{M}}[i, j] = \text{Fold}(\mathbf{u}, \widehat{\mathbf{M}}[*], j).$$

Thus, each coordinate of an encoded vector-matrix product can be checked using only the corresponding encoded column. Applying this identity to the Freivalds intermediates gives, for every $j \in [n]$, $\mathbf{x} = \mathbf{rA}$, then for every $j \in [n]$,

$$\widehat{\mathbf{x}}[j] = \text{Fold}(\mathbf{r}, \widehat{\mathbf{A}}[*], j), \quad \widehat{\mathbf{z}}[j] = \text{Fold}(\mathbf{r}, \widehat{\mathbf{C}}[*], j),$$

and

$$\widehat{\mathbf{y}}[j] = \text{Fold}(\mathbf{x}, \widehat{\mathbf{B}}[*], j).$$

The last equation alone, however, does not provide a proximity test for the committed encoding $\widehat{\mathbf{B}}$. The folding vector $\mathbf{x} = \mathbf{rA}$ depends on $\mathbf{A}$ and can be degenerate. For example, if $\mathbf{A} = 0$, then $\mathbf{x} = 0$ for every challenge $\mathbf{r}$, and the resulting $\mathbf{x}$ -fold reveals no information about whether $\widehat{\mathbf{B}}$ is close to a valid interleaved codeword. To obtain an independent proximity test for $\widehat{\mathbf{B}}$, the prover computes $\mathbf{b} = \mathbf{sB}$. By the same linearity property, $\widehat{\mathbf{b}}[j] = \text{Fold}(\mathbf{s}, \widehat{\mathbf{B}}[*], j)$ for every $j \in [n]$. Thus, the $\mathbf{r}$ -fold for $\mathbf{A}, \mathbf{C}$ and the $\mathbf{s}$ -fold for $\mathbf{B}$ tests the proximity of $\widehat{\mathbf{A}}, \widehat{\mathbf{C}}$ and $\widehat{\mathbf{B}}$, whereas the $\mathbf{x}$ -fold checks consistency of the claimed relation $\mathbf{y} = \mathbf{xB}$.

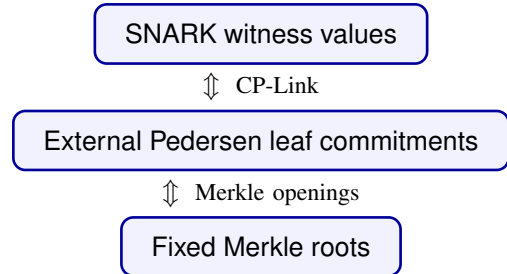
4.3. Sampled-Opening Layer

The sampled checks above are meaningful only if the values used inside the SNARK are bound to data fixed before the relevant challenges are sampled. LAMP achieves this through a sampled-opening layer. The prover first commits to the interleaved symbols, equivalently the encoded matrix columns, and fixes a Merkle root before the Freivalds challenge is derived. After the challenge is fixed, the prover computes the intermediate vectors, commits to their encoded symbols, and fixes a second Merkle root before the sampled positions are derived.

The SNARK circuit itself checks field equations: the sampled fold and equality checks, together with the code-consistency checks for the intermediate vectors. It does not verify Pedersen openings or Merkle paths internally. Instead, the commit-and-prove SNARK commits to the sampled witness blocks used by the circuit, and the prover supplies:

- Merkle openings showing that the external Pedersen commitments $\text{cm}_{ABC,j}$ and $\text{cm}_{XYZ,j}$ are authenticated under rt_{ABC} and rt_{XYZ} ;
- CP-Link proofs showing that the witness commitments from CP-SNARK and the external Pedersen commitments open to the corresponding sampled blocks.

Thus the verifier obtains a chain of consistency:



This separation is essential for efficiency. The SNARK checks only arithmetic relations on field elements, while Merkle authentication and commitment linking are verified outside the circuit.

4.4. Soundness Intuition

The soundness argument has three layers.

Freivalds soundness. If $\mathbf{AB} \neq \mathbf{C}$, then $\mathbf{D} = \mathbf{AB} - \mathbf{C}$ is nonzero. With the structured challenge $\mathbf{r} = (1, r, \dots, r^{k-1})$, the vector $\mathbf{rD}$ is nonzero except with probability at most $(k-1)/|\mathbb{F}|$, by Schwartz–Zippel. Therefore, except with this probability, at least one of the vector relations in the Freivalds reduction is false.

Proximity soundness.

The committed matrix encodings need not be exact codewords. If a committed encoding is τ -far from the interleaved code, the proximity-gap [37] property ensures, except with a small failure probability over the folding challenge, that its folded word remains τ -far from every valid codeword. Otherwise, if it is τ -close to an underlying codeword but the prover claims an inconsistent vector relation, the code distance δ guarantees that the folded word and the claimed encoding differ on at least a $(\delta - \tau)$ -fraction of positions. Since $\tau \leq \tau_{\text{UD}}$ implies $\delta - \tau \geq \tau$, an invalid claim leaves at least a τ -fraction of inconsistent positions in either case. Therefore, t random queries miss all inconsistencies with probability at most $(1 - \tau)^t$. The proximity-gap failure over the folding challenges is accounted for separately in the formal soundness analysis 6.

Commitment and linking soundness. The distance argument applies only to values fixed before the query positions are sampled. Merkle position binding prevents the prover from opening a root to two different leaves at the same position. Pedersen binding prevents a committed leaf from being opened to two different messages. CP-Link ensures that the values used by the SNARK are the same values authenticated by the external commitments. Therefore, except for the binding and linking failure probabilities, any accepting proof corresponds to fixed encoded objects that pass the sampled proximity checks.

Putting the layers together, a cheating prover must either break one of the commitment/linking mechanisms, violate SNARK soundness, hit the Freivalds failure event, or avoid all sampled disagreements in the encoded domain. The formal theorem in Section 6 makes this union bound explicit.

5. The LAMP Protocol

Section 4 reduced the original matrix-multiplication relation to the Freivalds relation and then to sampled encoded checks. This section formalizes the relation checked by LAMP. The important point is that LAMP is not a single arithmetic-circuit relation: the arithmetic checks are proved by an inner CP-SNARK, while Merkle membership and CP-Link are verified outside the circuit.

5.1. The Composite LAMP Relation

Let $\mathbf{A}, \mathbf{B}, \mathbf{C} \in \mathbb{F}^{k \times k}$, and let $\mathcal{C} : \mathbb{F}^k \rightarrow \mathbb{F}^n$ be a linear code with relative distance δ . The target matrix-multiplication relation is

$$\mathcal{R}_{\text{orig}} = \{(\mathbf{A}, \mathbf{B}, \mathbf{C}) : \mathbf{AB} = \mathbf{C}\}.$$

The verifier does not receive the original matrices as public input. Instead, the prover commits to row-wise encodings of the matrices and proves that the committed objects satisfy the multiplication relation.

The relation checked by LAMP is the composite relation

$$\mathcal{R}_{\text{LAMP}} = \mathcal{R}_{\text{LAMP}}^{\text{on}} \wedge \mathcal{R}_{\text{LAMP}}^{\text{off}}.$$

Here $\mathcal{R}_{\text{LAMP}}^{\text{on}}$ is the online relation proved by the CP-SNARK after the Freivalds challenge, query positions, and code-check points are sampled. The relation $\mathcal{R}_{\text{LAMP}}^{\text{off}}$ is the offline relation: it captures the values fixed by commitments before the online queries are known, and the verifier checks it through Merkle openings and CP-Link proofs. Sections 5.3 and 5.4 define these two relations explicitly. The circuit checks field equations on sampled encoded values, while the auxiliary verifier checks that those values are the ones bound by the pre-query commitments.

5.2. Commitment Layout and Public Challenges

The prover first encodes the matrices row-wise: $\hat{\mathbf{A}} = \text{Enc}(\mathbf{A})$, $\hat{\mathbf{B}} = \text{Enc}(\mathbf{B})$, $\hat{\mathbf{C}} = \text{Enc}(\mathbf{C})$. For each codeword position $j \in [n]$, it packs the three k -dimensional interleaved symbols, equivalently the three encoded columns, into $\mathbf{m}_{ABC,j} := (\hat{\mathbf{A}}[*], j) \parallel \hat{\mathbf{B}}[*], j \parallel \hat{\mathbf{C}}[*], j \in \mathbb{F}^{3k}$. Using a Pedersen commitment key ck_{ABC} , the prover samples $o_{ABC,j} \leftarrow \$ \mathbb{F}$ and computes $\text{cm}_{ABC,j} \leftarrow \text{Ped.Com}(\text{ck}_{ABC}, \mathbf{m}_{ABC,j}; o_{ABC,j})$. The matrix root is the Merkle commitment to these Pedersen commitments: $\text{rt}_{ABC} \leftarrow \text{MT.Com}((\text{cm}_{ABC,j})_{j \in [n]}; \perp)$. Here and below, $\perp$ indicates that no additional randomness is used in the Merkle tree, since the leaves are Pedersen commitments that already provide hiding. The root rt_{ABC} is fixed before the Freivalds challenge is derived.

After receiving rt_{ABC} , the verifier samples $r, s \leftarrow \$ \mathbb{F}$. The prover sets $\mathbf{r} = (1, r, r^2, \dots, r^{k-1})$, $\mathbf{s} = (1, s, \dots, s^{k-1})$, and computes $\mathbf{x} = \mathbf{rA}$, $\mathbf{y} = \mathbf{xB}$, $\mathbf{z} = \mathbf{rC}$, $\mathbf{b} = \mathbf{sB}$. It then encodes the intermediate vectors: $\hat{\mathbf{x}} = \text{Enc}(\mathbf{x})$, $\hat{\mathbf{y}} = \text{Enc}(\mathbf{y})$, $\hat{\mathbf{z}} = \text{Enc}(\mathbf{z})$, $\hat{\mathbf{b}} = \text{Enc}(\mathbf{b})$. The encoded intermediate vectors form a 4-fold interleaved codeword. For each $j \in [n]$, the prover packs its j -th interleaved symbol into $\mathbf{m}_{XYZ,j} := (\hat{\mathbf{x}}[j] \parallel \hat{\mathbf{y}}[j] \parallel \hat{\mathbf{z}}[j] \parallel \hat{\mathbf{b}}[j]) \in \mathbb{F}^4$. Using a Pedersen commitment key ck_{XYZ} , the prover samples $o_{XYZ,j} \leftarrow \$ \mathbb{F}$, computes $\text{cm}_{XYZ,j} \leftarrow \text{Ped.Com}(\text{ck}_{XYZ}, \mathbf{m}_{XYZ,j}; o_{XYZ,j})$, and sends the vector-symbol root $\text{rt}_{XYZ} \leftarrow \text{MT.Com}((\text{cm}_{XYZ,j})_{j \in [n]}; \perp)$ to the verifier.

Only after receiving rt_{XYZ} does the verifier sample the query tuple $I = (j_1, \dots, j_t) \leftarrow \$ [n]^t$. It also samples public code-check points $\alpha_x, \alpha_y, \alpha_z, \alpha_b \leftarrow \$ \mathbb{F} \setminus D$, where D is the code evaluation domain. These points are used by the circuit to check that $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}, \hat{\mathbf{b}}$ are encodings of the claimed intermediate vectors. For the Reed–Solomon code used in our implementation, this check is the barycentric out-of-domain consistency test recalled in Appendix D.

5.3. The Online Relation

The online relation is the arithmetic relation proved inside the inner CP-SNARK. Its role is not to verify Merkle paths or Pedersen openings. Instead, it checks only the field equations that certify the sampled encoded positions are consistent with the Freivalds reduction.

The public statement contains the Merkle roots and verifier-sampled challenges:

$$\mathbf{x} = (\mathbf{rt}_{ABC}, \mathbf{rt}_{XYZ}, r, I, \alpha_x, \alpha_y, \alpha_z, \alpha_b).$$

The roots are included in the statement to bind the CP-SNARK proof to the same transcript and offline checks; the online field equations themselves do not evaluate Merkle paths.

The complete SNARK witness is $\mathbf{w} = (\mathbf{u}, \omega)$, where the committed part of the witness is

$$\mathbf{u} = ((\mathbf{m}_{ABC,j})_{j \in I}, (\mathbf{m}_{XYZ,j})_{j \in I}),$$

and the remaining private witness is $\omega = (\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{b}, \hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}, \hat{\mathbf{b}})$. The CP-SNARK treats $\mathbf{u}$ as the committed part of the witness. The prover first computes the SNARK-side witness commitments $\widetilde{\mathbf{cm}}_{ABC}$ and $\widetilde{\mathbf{cm}}_{XYZ}$ using $\Pi_{cp}.Com$, and then proves the online relation with witness $\mathbf{w} = (\mathbf{u}, \omega)$. These witness commitments are not checked against the Merkle roots inside the circuit. Instead, they are connected to the externally committed Merkle leaves by the offline relation in Section 5.4. For the CP-Link statements below, let $\mathbf{ck}_{cp,T}$ denote the public commitment key used by the CP-SNARK witness commitment $\widetilde{\mathbf{cm}}_T$, for $T \in \{ABC, XYZ\}$. Write $\text{EncCheck}_C(\hat{\mathbf{v}}, \mathbf{v}; \alpha) = 1$ when the out-of-domain code-consistency check accepts that $\hat{\mathbf{v}}$ is consistent with $\text{Enc}(\mathbf{v})$ at the public point α . The online relation is defined in Figure 1.

```

o  $\mathcal{R}_{\text{LAMP}}^{\text{on}}(\mathbf{x}; \mathbf{w}) :$ 
  parse
     $\mathbf{x} = (\mathbf{rt}_{ABC}, \mathbf{rt}_{XYZ}, r, I, \alpha_x, \alpha_y, \alpha_z, \alpha_b)$ 
     $\mathbf{w} = (\mathbf{u}, \omega)$ 
     $\mathbf{u} = ((\mathbf{m}_{ABC,j})_{j \in I}, (\mathbf{m}_{XYZ,j})_{j \in I})$ 
     $\omega = (\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{b}, \hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}, \hat{\mathbf{b}})$ 
   $\mathbf{r} \leftarrow (1, r, \dots, r^{k-1})$ 
   $\mathbf{s} \leftarrow (1, s, \dots, s^{k-1})$ 
   $\text{EncCheck}_C(\hat{\mathbf{x}}, \mathbf{x}; \alpha_x) = 1$ 
   $\text{EncCheck}_C(\hat{\mathbf{y}}, \mathbf{y}; \alpha_y) = 1$ 
   $\text{EncCheck}_C(\hat{\mathbf{z}}, \mathbf{z}; \alpha_z) = 1$ 
   $\text{EncCheck}_C(\hat{\mathbf{b}}, \mathbf{b}; \alpha_b) = 1$ 
   $\forall j \in I :$ 
    parse  $\mathbf{m}_{ABC,j}$  as  $(\hat{\mathbf{A}}[*,j] \parallel \hat{\mathbf{B}}[*,j] \parallel \hat{\mathbf{C}}[*,j])$ 
    parse  $\mathbf{m}_{XYZ,j}$  as  $(\hat{\mathbf{x}}[j] \parallel \hat{\mathbf{y}}[j] \parallel \hat{\mathbf{z}}[j] \parallel \hat{\mathbf{b}}[j])$ 
     $\hat{\mathbf{x}}[j] = \text{Fold}(\mathbf{r}, \hat{\mathbf{A}}[*,j])$ 
     $\hat{\mathbf{y}}[j] = \text{Fold}(\mathbf{x}, \hat{\mathbf{B}}[*,j])$ 
     $\hat{\mathbf{z}}[j] = \text{Fold}(\mathbf{r}, \hat{\mathbf{C}}[*,j])$ 
     $\hat{\mathbf{b}}[j] = \text{Fold}(\mathbf{s}, \hat{\mathbf{B}}[*,j])$ 
     $\hat{\mathbf{y}}[j] = \hat{\mathbf{z}}[j]$ 

```

Figure 1. The Online Relation

The first four checks certify that $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$ are valid encodings of the claimed intermediate vectors $\mathbf{x}, \mathbf{y}, \mathbf{z}$. The remaining checks are local equations on the sampled codeword positions. For each queried position j , the circuit unpacks the sampled matrix-column block $\mathbf{m}_{ABC,j}$ and the sampled vector-symbol block $\mathbf{m}_{XYZ,j}$, and then verifies the four fold relations together with the final equality $\hat{\mathbf{y}}[j] = \hat{\mathbf{z}}[j]$.

Thus, the online relation only checks arithmetic consistency of the sampled values. It does not by itself guarantee that these sampled values came from the pre-query Merkle roots $\mathbf{rt}_{ABC}$ and $\mathbf{rt}_{XYZ}$. That binding is provided by the offline relation below.

For compactness, Figure 2 writes the CP-Link statement for $T \in \{ABC, XYZ\}$ as

$$\mathbf{x}_{\text{link}}^T := ((\mathbf{ck}_{cp,T}, \widetilde{\mathbf{cm}}_T), (\mathbf{ck}_T, \mathbf{cm}_{T,j})_{j \in I}),$$

where $\mathbf{ck}_{cp,T}$ is the witness-commitment key used by the CP-SNARK commitment $\widetilde{\mathbf{cm}}_T$, and $\mathbf{ck}_T$ is the external Pedersen commitment key for the Merkle leaves of type T .

5.4. The Offline Relation

The offline relation connects the sampled witnesses used by the CP-SNARK to the commitments that were

$$\begin{aligned}
& \circ \mathcal{R}_{\text{LAMP}}^{\text{off}} \left(\text{rt}_{ABC}, \text{rt}_{XYZ}, I, \widetilde{\text{cm}}_{ABC}, \widetilde{\text{cm}}_{XYZ}, \right. \\
& \quad \pi_{\text{link}}^{ABC}, \pi_{\text{link}}^{XYZ}, \\
& \quad \left. (\text{cm}_{ABC,j}, \pi_{\text{mem},j}^{ABC})_{j \in I}, (\text{cm}_{XYZ,j}, \pi_{\text{mem},j}^{XYZ})_{j \in I} \right) : \\
& \quad \forall j \in I : \\
& \quad \quad \text{MT.Verify}(\text{rt}_{ABC}, j, \text{cm}_{ABC,j}, \pi_{\text{mem},j}^{ABC}) = 1 \\
& \quad \quad \text{MT.Verify}(\text{rt}_{XYZ}, j, \text{cm}_{XYZ,j}, \pi_{\text{mem},j}^{XYZ}) = 1 \\
& \quad \Pi_{\text{link}} \cdot \text{Verify}(\text{pp}_{\text{link}}, \mathbf{x}_{\text{link}}^{ABC}, \pi_{\text{link}}^{ABC}) = 1 \\
& \quad \Pi_{\text{link}} \cdot \text{Verify}(\text{pp}_{\text{link}}, \mathbf{x}_{\text{link}}^{XYZ}, \pi_{\text{link}}^{XYZ}) = 1
\end{aligned}$$

Figure 2. The Offline Relation

fixed before the query positions were sampled. The term “offline” indicates that these checks are performed outside the SNARK circuit. In our system, it also reflects that the committed data are fixed before the online sampled checks are carried out.

For each queried position $j \in I$, the verifier receives Merkle openings for the external Pedersen leaves $\text{cm}_{ABC,j}$ and $\text{cm}_{XYZ,j}$. The Merkle openings certify that these leaves are contained in the roots rt_{ABC} and rt_{XYZ} , respectively. In addition, the verifier checks CP-Link proofs connecting the CP-SNARK witness commitments $\widetilde{\text{cm}}_{ABC}$ and $\widetilde{\text{cm}}_{XYZ}$ to the corresponding external Pedersen leaves.

The CP-Link relation is interpreted blockwise. Each block $\mathbf{m}_{T,j}$ is the serialized interleaved symbol opened at position j for the corresponding committed object T . For $T \in \{ABC, XYZ\}$, define the ordered sampled message vector

$$\mathbf{M}_{T,I} := (\mathbf{m}_{T,j_1} \parallel \mathbf{m}_{T,j_2} \parallel \cdots \parallel \mathbf{m}_{T,j_t}), \quad I = (j_1, \dots, j_t).$$

The SNARK-side witness commitment $\widetilde{\text{cm}}_T$ commits to the concatenated sampled witness $\mathbf{M}_{T,I}$, while each external Pedersen leaf cm_{T,j_ℓ} commits to the corresponding block $\mathbf{m}_{T,j_\ell}$. The public CP-Link statement includes the corresponding commitment keys:

$$\mathbf{x}_{\text{link}}^T := ((\text{ck}_{\text{cp},T}, \widetilde{\text{cm}}_T), (\text{ck}_T, \text{cm}_{T,j_\ell})_{\ell \in [t]}).$$

Therefore, the CP-Link proof for T proves the existence of openings such that $\widetilde{\text{cm}}_T = \Pi_{\text{cp}} \cdot \text{Com}(\text{pp}, \mathbf{M}_{T,I}; \tilde{o}_T)$, and, for every $\ell \in [t]$, $\text{cm}_{T,j_\ell} = \text{Ped} \cdot \text{Com}(\text{ck}_T, \mathbf{m}_{T,j_\ell}; o_{T,j_\ell})$. This blockwise interpretation is important: the CP-Link proof claims that the ordered blocks committed inside the CP-SNARK are exactly the same as the ordered blocks committed by the corresponding external Pedersen leaves.

The offline relation is defined in Figure 2.

By combining the Merkle membership checks with the CP-Link checks, we obtain the consistency chain described in Section 4.3. For each $T \in \{ABC, XYZ\}$,

$$\begin{array}{ccc}
\widetilde{\text{cm}}_T & \xleftrightarrow{\pi_{\text{link}}^T} & (\text{cm}_{T,j})_{j \in I} \\
(\text{cm}_{T,j})_{j \in I} & \xleftrightarrow{(\pi_{\text{mem},j}^T)_{j \in I}} & \text{rt}_T
\end{array}$$

Here $T = ABC$ uses the packed sampled matrix-column messages $(\mathbf{m}_{ABC,j})_{j \in I}$ and the root rt_{ABC} , while $T =$

XYZ uses the packed sampled vector-symbol messages $(\mathbf{m}_{XYZ,j})_{j \in I}$ and the root rt_{XYZ} .

The upper line states that the SNARK-side committed witness and the opened external Pedersen leaves contain the same ordered sampled blocks. The lower line states that those external leaves are authenticated under the corresponding pre-query Merkle root. Therefore, once both $\mathcal{R}_{\text{LAMP}}^{\text{on}}$ and $\mathcal{R}_{\text{LAMP}}^{\text{off}}$ hold, the verifier knows that the arithmetic checks performed inside the CP-SNARK were applied to values already bound by rt_{ABC} and rt_{XYZ} . This separation is what allows the SNARK circuit to avoid in-circuit Pedersen verification and in-circuit Merkle path verification.

5.5. Protocol Algorithms and Complexity

Figure 3 presents the public-coin version of LAMP. The protocol can be made non-interactive via the Fiat-Shamir transform, by deriving r, s after rt_{ABC} and deriving $I, \alpha_x, \alpha_y, \alpha_z, \alpha_b$ after rt_{XYZ} .

The proof consists of the CP-SNARK proof, the sampled leaf commitments and Merkle openings, and the CP-Link proofs. The sampled matrix-column blocks and vector-symbol blocks are private CP-SNARK witnesses; they are not serialized as public proof material. The public statement contains the roots $\text{rt}_{ABC}, \text{rt}_{XYZ}$ and the verifier-sampled challenges.

For every $T \in \mathcal{T} = \{ABC, XYZ\}$, Figure 3 uses the following CP-Link statement and witness:

$$\begin{aligned}
\mathbf{x}_{\text{link}}^T &:= ((\text{ck}_{\text{cp},T}, \widetilde{\text{cm}}_T), (\text{ck}_T, \text{cm}_{T,j})_{j \in I}), \\
\mathbf{w}_{\text{link}}^T &:= ((\mathbf{m}_{T,j})_{j \in I}, \tilde{o}_T, (o_{T,j})_{j \in I}).
\end{aligned}$$

Thus the CP-Link API is used with the public linking parameters, the commitment keys and commitments as the statement, and the sampled messages together with their opening randomness as the witness.

Cost model. We report the cost of the LAMP. The computation of the claimed output matrix $C = AB$ is outside this cost model. We do count the work performed by the LAMPprover after the matrices are available, including encoding, commitment generation, computation of the Freivalds intermediate witnesses, sampled Merkle openings, CP-SNARK proving, and CP-Link proving.

Let $E_{\text{Enc}}(k, n)$ denote cost of the corresponding encoding phase. Let $E_{\text{code}} = E_{\text{code}}(k, n)$ denote the constraint cost of the intermediate code-consistency checks; for the Reed-Solomon check in Appendix D, this is $\mathcal{O}(k + n)$ after domain-dependent coefficients are precomputed. We write $\mathbb{G}$ for group operations used by Pedersen commitments, $\mathbb{F}$ for field operations, and $\mathbb{H}$ for hash evaluations. Let $E_{\text{snark}}^{\text{P}}$ and $E_{\text{snark}}^{\text{V}}$ denote the CP-SNARK proving and verification costs, respectively. Let $E_{\text{link}}^{\text{P}}(k, t)$ and $E_{\text{link}}^{\text{V}}(k, t)$ denote the prover and verifier costs, respectively, of the selected CP-Link construction. For the QALink instantiation in Appendix E, each of the two link statements has $q = t$

external commitments, so $E_{\text{link}}^{\mathcal{V}}(k, t)$ includes $2(t+2)$ Miller-loop inputs, batched into one final exponentiation, plus $\mathcal{O}(t)$ group scalar multiplications for batching.

Table 2 summarizes the asymptotic costs at the level of the main protocol components.

Component	Prover	Verifier
ABC encoding	$E_{\text{Enc}}(k, n)$	—
ABC commit	$\mathcal{O}(3nk)\mathbb{G} + \mathcal{O}(n)\mathbb{H}$	—
Intermediate	$\mathcal{O}(k^2)\mathbb{F}$	—
Intermediate encoding	$E_{\text{Enc}}(k, n)$	—
Intermediate commit	$\mathcal{O}(n)\mathbb{G} + \mathcal{O}(n)\mathbb{H}$	—
Merkle openings	$\mathcal{O}(t \log n)\mathbb{H}$	$\mathcal{O}(t \log n)\mathbb{H}$
CP-SNARK	$\mathcal{O}(tk) + E_{\text{code}} \text{ constraints}$	$E_{\text{snark}}^{\mathcal{V}}$
CP-Link	$E_{\text{link}}^{\mathcal{P}}(k, t)$	$E_{\text{link}}^{\mathcal{V}}(k, t)$

TABLE 2. ASYMPTOTIC COSTS FOR LAMP.

Prover work. The prover first encodes the three input matrices and commits to the packed encoded matrix columns. The ABC commitment phase consists of n Pedersen commitments to blocks of length $3k$, followed by a Merkle root over the resulting leaves. After the Freivalds challenge r is sampled, the prover computes the intermediate witnesses $\mathbf{x} = \mathbf{rA}, \mathbf{y} = \mathbf{xB}, \mathbf{z} = \mathbf{rC}, \mathbf{b} = \mathbf{sB}$ which costs $\mathcal{O}(k^2)$ field operations. This is distinct from the excluded native computation of $C = AB$. The prover then encodes $\mathbf{x}, \mathbf{y}, \mathbf{z}$, commits to their encoded symbols, opens the sampled Merkle paths, proves the online CP-SNARK relation, and produces the CP-Link proofs.

For each queried position $j \in I$, the CP-SNARK circuit checks **four** length- k fold equations:

$$\begin{aligned} &\text{Fold}(\mathbf{r}, \widehat{\mathbf{A}}[*, j]), \text{Fold}(\mathbf{x}, \widehat{\mathbf{B}}[*, j]), \\ &\text{Fold}(\mathbf{r}, \widehat{\mathbf{C}}[*, j]), \text{Fold}(\mathbf{b}, \widehat{\mathbf{B}}[*, j]). \end{aligned}$$

Thus the sampled arithmetic checks contribute $\mathcal{O}(tk)$ constraints.

The intermediate-vector code-consistency checks add E_{code} constraints. For the fixed-rate Reed–Solomon setting used in the experiments, $n = 2k$, so this extra cost is linear in k and the total online relation remains linear in k for fixed t .

Verifier work. The verifier does not encode matrices, compute intermediate vectors, or build commitment roots. Its work consists of CP-SNARK verification, verification of the sampled Merkle authentication paths, and CP-Link verification. Therefore, using the notation above, the verifier cost is $E_{\text{snark}}^{\mathcal{V}} + \mathcal{O}(t \log n)\mathbb{H} + E_{\text{link}}^{\mathcal{V}}(k, t)$, where the QALink verifier contributes $\mathcal{O}(t)$ pairing terms.

Comparison. A naive SNARK arithmetization of matrix multiplication costs $\mathcal{O}(k^3)$ constraints, while a Freivalds-based SNARK baseline computes the vector-matrix products inside the circuit and costs $\mathcal{O}(k^2)$ constraints. LAMP moves

the vector-matrix products to native prover work and checks sampled encoded-column relations plus intermediate code-consistency equations inside the CP-SNARK. As a result, the main circuit constraint count is $\mathcal{O}(tk + E_{\text{code}})$, which is linear in k for the fixed-rate code used in our implementation.

6. Security Analysis

We prove perfect completeness and computational soundness for the public-coin interactive LAMP scheme. The proof is stated for the protocol in which the verifier samples the Freivalds challenge, the query positions, and the code-check points.

Statement and scope. Section 5 defines the checked relation as the composite relation

$$\mathcal{R}_{\text{LAMP}} = \mathcal{R}_{\text{LAMP}}^{\text{on}} \wedge \mathcal{R}_{\text{LAMP}}^{\text{off}}.$$

The online component $\mathcal{R}_{\text{LAMP}}^{\text{on}}$ is enforced by the CP-SNARK, while the offline binding component $\mathcal{R}_{\text{LAMP}}^{\text{off}}$ is enforced by sampled Merkle membership checks and CP-Link proofs. This section proves that satisfying both components implies soundness for the committed matrix-multiplication statement.

The theorem is relative to a committed input root rt_{ABC} . Namely, rt_{ABC} binds, before the Freivalds challenge is sampled, to row-wise encodings

$$\widehat{\mathbf{A}} = \text{Enc}(\mathbf{A}), \quad \widehat{\mathbf{B}} = \text{Enc}(\mathbf{B}), \quad \widehat{\mathbf{C}} = \text{Enc}(\mathbf{C})$$

of fixed matrices $\mathbf{A}, \mathbf{B}, \mathbf{C} \in \mathbb{F}^{k \times k}$. Equivalently, rt_{ABC} fixes the corresponding interleaved codewords, so each sampled index j refers to a fixed encoded column before the query tuple is known. The multiplication statement is false when these matrices satisfy $\mathbf{AB} \neq \mathbf{C}$.

After the verifier samples the Freivalds challenge, the prover commits to the intermediate-vector messages $\mathbf{x}, \mathbf{y}, \mathbf{z}$ and to their encoded views before the verifier samples the query tuple I and the code-check points. This ordering is the timing property needed by the proof: all values tested by the sampled proximity checks are fixed before the sampled positions are known.

Definition 6.1 (LAMP backend assumptions). The backend used by LAMP is assumed to satisfy the following properties.

- *CP-SNARK soundness.* Any accepting CP-SNARK proof for $\mathcal{R}_{\text{LAMP}}^{\text{on}}$ has a valid online-checking witness bound to the public SNARK-side witness commitments $\widetilde{\text{cm}}_{ABC}$ and $\widetilde{\text{cm}}_{XYZ}$, generated by $\Pi_{\text{cp}}.\text{Com}$, except with probability at most $\epsilon_{\text{cp}}(\lambda)$.
- *Opening binding.* Accepted Merkle openings fix unique Pedersen leaves at the queried positions, and accepted Pedersen leaves are binding to unique packed messages, except with probability at most $\epsilon_{\text{open}}(\lambda)$.
- *CP-Link consistency.* Accepted CP-Link proofs imply that the SNARK-side committed sampled blocks and the opened external Pedersen leaves contain the

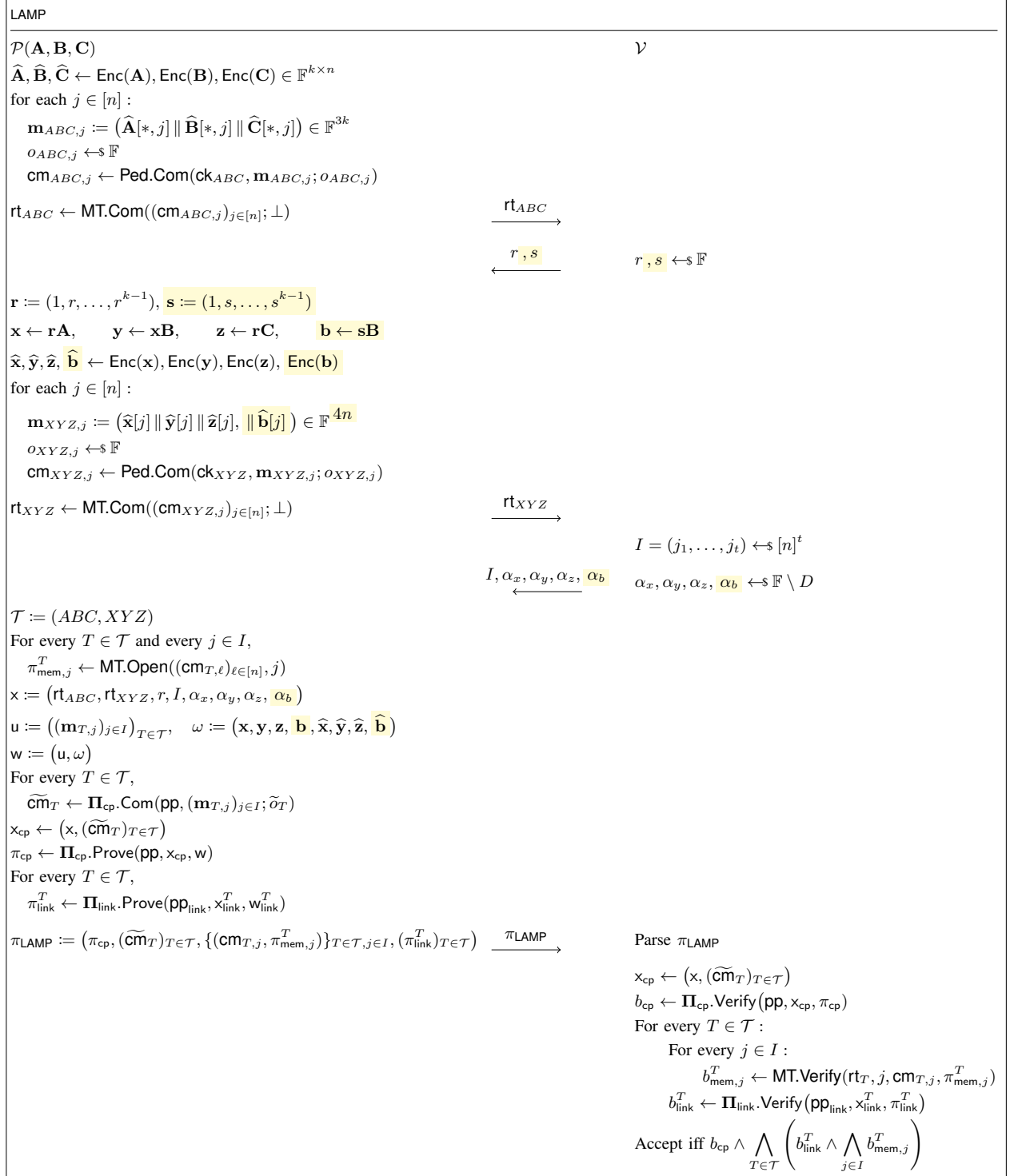


Figure 3. The LAMP protocol

same packed messages, except with probability at most $\epsilon_{\text{link}}(\lambda)$.

- **Code-consistency soundness.** The probabilistic encoding checks for $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$ fail with probability at most $\epsilon_{\text{code}}(\lambda)$. For the Reed–Solomon consistency check used by LAMP, if each code-check point is sampled uniformly from outside the evaluation domain after the intermediate commitments are fixed, then

$$\epsilon_{\text{code}} \leq \frac{3(n-1)}{|\mathbb{F}| - n}.$$

Appendix D gives the corresponding barycentric Reed–Solomon check and its soundness bound.

Theorem 6.2 (Completeness, soundness, and zero knowledge). *Assume that the input root $\mathbf{rt}_{ABC}$ is fixed before the Freivalds challenge and binds to valid row-wise encodings of fixed matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}$, as described above. Assume also that the backend satisfies Definition 6.1. Then the interactive LAMP scheme satisfies:*

- (i) **Perfect completeness.** *If $\mathbf{AB} = \mathbf{C}$, the honest prover is accepted with probability one.*
- (ii) **Interactive computational soundness.** *For every PPT prover $\mathcal{P}^*$, if the matrices bound by $\mathbf{rt}_{ABC}$ satisfy $\mathbf{AB} \neq \mathbf{C}$, then*

$$\Pr[\text{Acc}] \leq \epsilon_{\text{cp}}(\lambda) + \epsilon_{\text{open}}(\lambda) + \epsilon_{\text{link}}(\lambda) + \epsilon_{\text{code}}(\lambda) + \frac{k-1}{|\mathbb{F}|} + 4(1-\delta)^t.$$

Here $t = |I|$ is the number of sampled codeword positions and δ is the relative distance of the code $\mathcal{C}$.

- (iii) **Zero knowledge.** *Assume that Π_{cp} is zero knowledge, Π_{link} is witness zero knowledge, and the Pedersen commitments are hiding. Let $\mathbf{x}_{\text{LAMP}}$ denote the public transcript data. Then there exists a simulator Sim_{LAMP} such that, for every PPT distinguisher $\mathcal{D}$,*

$$\text{View}_{\text{LAMP}}^{\mathcal{P}, \mathcal{V}}(\mathbf{x}_{\text{LAMP}}, \mathbf{w}) \approx_c \text{Sim}_{\text{LAMP}}(\mathbf{x}_{\text{LAMP}}).$$

Proof sketch. Completeness follows from the honest choice

$$\mathbf{x} = \mathbf{rA}, \quad \mathbf{y} = \mathbf{xB}, \quad \mathbf{z} = \mathbf{rC}.$$

If $\mathbf{AB} = \mathbf{C}$, then $\mathbf{y} = \mathbf{z}$. By linearity of $\mathcal{C}$, the sampled fold equations in $\mathcal{R}_{\text{LAMP}}^{\text{on}}$ hold at every queried position, and the honestly generated Merkle openings and CP-Link proofs satisfy $\mathcal{R}_{\text{LAMP}}^{\text{off}}$.

For soundness, condition on the complement of the CP-SNARK, opening, CP-Link, and code-consistency failure events. These events contribute $\epsilon_{\text{cp}} + \epsilon_{\text{open}} + \epsilon_{\text{link}} + \epsilon_{\text{code}}$. Under this conditioning, every sampled value used inside the CP-SNARK is the value opened from the corresponding pre-query commitment, and the committed intermediate encoded views are valid encodings of their claimed messages.

If $\mathbf{AB} \neq \mathbf{C}$, then for $\mathbf{D} = \mathbf{AB} - \mathbf{C} \neq 0$, the structured Freivalds challenge $\mathbf{r} = (1, r, \dots, r^{k-1})$ satisfies $\mathbf{rD} = 0$ with probability at most $(k-1)/|\mathbb{F}|$ by Schwartz–Zippel. Conditioned on $\mathbf{rD} \neq 0$, at least one of the four global vector relations underlying the online checks is false. The

two corresponding codewords then disagree on at least a δ -fraction of positions, so t independent queries miss the disagreement with probability at most $(1-\delta)^t$. Union-bounding over the four checked relations gives $4(1-\delta)^t$.

Combining these events yields the stated bound. Appendix C gives the detailed bad-event decomposition and the Fiat–Shamir variant.

Zero knowledge follows by a standard hybrid argument. The CP-SNARK proof is simulated using the zero-knowledge simulator of Π_{cp} , and each CP-Link proof is simulated using the witness-zero-knowledge simulator of Π_{link} . The remaining public data consists only of Pedersen commitments, Merkle roots, and Merkle authentication paths. By Pedersen hiding, these commitments reveal no information about the committed witness values; the Merkle layer only authenticates the commitments as leaves and does not open the underlying values. Hence the simulated transcript is computationally indistinguishable from a real execution. $\square$

Remark 3 (Fiat–Shamir). The theorem above analyzes the public-coin interactive protocol. A non-interactive variant can derive r , I , and the code-check points from domain-separated transcript hashes via the Fiat–Shamir transform. Proving that variant adds the standard random oracle model loss and is orthogonal to the interactive soundness statement above.

7. Implementation and Evaluation

We implement LAMP and evaluate its performance. The implementation follows the construction in Section 5. We instantiate the CP-SNARK with Groth16, using the commit-and-prove interface [10], [34]. For CP-Link, we use a pairing-based QA-NIZK proof [38]; the concrete instantiation used in our implementation is given in Appendix E. The implementation is available at <https://anonymous.4open.science/r/LAMP-38E3/>.

Our evaluation is organized around four questions. First, does the number of constraints of LAMP scale linearly with the matrix dimension, as predicted by Section 5.5? Second, which components dominate proving time, proof size, and verification time? Third, how does LAMP behave when proving multiple independent claims $\mathbf{A}_i \mathbf{B}_i = \mathbf{C}_i$ in a batch, rather than a single product $\mathbf{AB} = \mathbf{C}$? Fourth, does the advantage remain useful on a neural-network workload?

Experimental setup. We implement both LAMP and the Freivalds baseline in Go using gnark, with BN254 as the elliptic curve. All experiments were run on a Linux server with an AMD EPYC 7B13 CPU with 32 cores and 254GB of memory. We use a Reed–Solomon code of rate $\rho = 1/2$ and set the number of sampled positions to $t = 128$.

7.1. Comparison with a Freivalds Baseline

We compare LAMP with a Freivalds-based baseline implemented as a SNARK circuit. Table 3 reports the number

k	Constraints		Prove Time		Verify Time		Proof Size	
	LAMP	Freivalds	LAMP	Freivalds	LAMP	Freivalds	LAMP	Freivalds
2^7	54,069	49,610	0.76 s	0.58 s	0.06 s	1 ms	22,852 B	
2^8	105,915	197,194	1.50 s	2.03 s	0.06 s	1 ms	29,380 B	
2^9	209,995	788,810	3.08 s	6.89 s	0.06 s	1 ms	36,292 B	196 B
2^{10}	418,149	3,156,298	7.11 s	25.63 s	0.06 s	2 ms	43,332 B	
2^{11}	834,075	12,622,154	16.80 s	102.07 s	0.06 s	2 ms	51,076 B	
2^{12}	1,666,315	50,495,818	48.06 s	405.04 s	0.06 s	2 ms	58,564 B	
2^{13}	3,330,789	–	150.24 s	–	0.06 s	–	66,756 B	–

TABLE 3. LAMP COMPARED WITH A FREIVALDS BASELINE.

of constraints, proving time, verification time, and proof size for both systems.

For small dimensions, the fixed cost of $\mathcal{R}_{\text{LAMP}}^{\text{on}}$ dominates. In particular, at $k = 2^7$, LAMP uses slightly more constraints than the baseline. Starting from $k = 2^8$, however, the asymptotic advantage of LAMP becomes visible. As k doubles, the Freivalds baseline grows by roughly $4\times$, whereas LAMP grows by roughly $2\times$. This behavior matches the expected gap between the $\mathcal{O}(k^2)$ vector-matrix computation in the baseline and the $\mathcal{O}(tk)$ sampled fold check in LAMP. At $k = 4,096$, LAMP reduces the number of constraints from 50,495,818 to 1,666,315, corresponding to a $30.30\times$ reduction. The proving time also decreases from 405.04 s to 48.06 s. At $k = 8,192$, the Freivalds baseline runs out of memory, while LAMP still completes with 3,330,789 constraints.

The main cost of LAMP appears in verification and proof size. Verification takes about 0.06 s for LAMP, compared with 1–2 ms for the Freivalds baseline. This additional overhead comes from verifying Merkle openings and the pairing-based CP-Link proofs in addition to the Groth16 proof. In the QA-Link instantiation, CP-Link verification is the dominant part of this overhead because the verifier evaluates the sampled link equations as a multi-pairing check. Across all measured matrix dimensions, verification stays at about 0.06 s, while the prover-side savings become substantial for large matrix dimensions.

7.2. LAMP Component Breakdown

Table 4 breaks down the measured cost of LAMP by component. Commit time includes both Pedersen leaf commitments and Merkle root generation. For small dimensions, SNARK proving is the main bottleneck. As k grows, however, commitment generation for the ABC and XYZ objects becomes dominant. For example, at $k = 4,096$, commitment generation takes 34.13 s in total, while Groth16 proving takes 13.60 s and CP-Link proving takes 0.33 s. On the verifier side, Groth16 and Merkle verification remain at a few milliseconds, while the pairing-based QA-Link check takes about 0.05–0.06 s. Thus total verification remains about 0.06 s for all measured matrix dimensions.

7.3. Batching Matrix-Multiplication Claims

We also evaluate a batched setting with multiple independent claims $\mathbf{A}_i\mathbf{B}_i = \mathbf{C}_i$. The number of claims is the number of matrix products proved together. For each codeword position, the prover packs all interleaved symbols of $\mathbf{A}_i, \mathbf{B}_i, \mathbf{C}_i$ into a single Pedersen leaf and builds one Merkle tree. Therefore, the number of Merkle membership proofs does not grow with the number of claims.

Table 5 reports the component costs at fixed $k = 2^7$. As expected, the constraint count and Groth16 proving time grow with the number of checked matrix products. In contrast, the Merkle proof size stays around 22 KB across all claim counts. This shows that, once the batched matrix leaves are committed together, the sampled Merkle membership proof does not scale with the number of matrix products. The small variation in Merkle proof size comes from the multi-opening implementation: the number of shared authentication nodes can vary slightly depending on the sampled index set. Verification time remains about 0.06–0.07 s across the batch-size range, because the number of sampled positions and the verification procedure are unchanged.

7.4. GPT-2 Workload

Table 6 evaluates LAMP as a verifiable AI application: the matrix multiplications in a single GPT-2 layer with sequence length 2^{10} . This workload consists of several matrix multiplications arising from the attention and MLP components of the layer, with different matrix shapes and structures. LAMP reduces the constraint count from 76,807,671 to 33,427,908, a $2.30\times$ reduction, and reduces proving time from 408.69 s to 230.82 s, a $1.77\times$ improvement. The gain is smaller than in the square-matrix benchmark because a GPT-2 layer decomposes into several rectangular matrix multiplications and smaller matrix products inside each attention head. Since these products have different dimensions, the sampled opening layer cannot easily amortize all products through a single shared leaf layout. This effect also appears in verifier-side costs. The Freivalds baseline verifies in 0.49 s, whereas LAMP uses more commitments and therefore requires the verifier to check more CP-Link proofs and Merkle membership proofs. As a result, verification takes 6.52 s for LAMP, and the proof size increases to 3,342,724

k	Commit Time		Prove Time			Verify Time			Proof Size		
	ABC	XYZ	Merkle	Groth16	CP-Link	Groth16	Merkle	CP-Link	Merkle	Groth16	CP-Link
2^7	0.09 s	0.01 s	1 ms	0.63 s	0.02 s	2 ms	1 ms	0.05 s	22,464 B		
2^8	0.25 s	0.02 s	1 ms	1.18 s	0.05 s	2 ms	1 ms	0.05 s	28,992 B		
2^9	0.87 s	0.03 s	1 ms	2.11 s	0.07 s	2 ms	1 ms	0.06 s	35,904 B		
2^{10}	3.09 s	0.13 s	1 ms	3.78 s	0.11 s	2 ms	1 ms	0.05 s	42,944 B	260 B	128 B
2^{11}	9.15 s	0.45 s	1 ms	7.03 s	0.17 s	2 ms	1 ms	0.05 s	50,688 B		
2^{12}	32.32 s	1.81 s	1 ms	13.60 s	0.33 s	2 ms	2 ms	0.05 s	58,176 B		
2^{13}	116.78 s	6.23 s	1 ms	26.51 s	0.72 s	2 ms	2 ms	0.05 s	66,368 B		

TABLE 4. COMPONENT-LEVEL MEASUREMENTS FOR LAMP.

Claims	Constraints	Commit Time		Prove Time			Proof Size		
		ABC	XYZ	Merkle	Groth16	CP-Link	Merkle	Groth16	CP-Link
1	54,069	0.11 s	0.01 s	1 ms	0.65 s	0.02 s	22,592 B		
2	105,349	0.15 s	0.01 s	1 ms	1.18 s	0.04 s	22,528 B		
3	156,629	0.19 s	0.02 s	1 ms	1.69 s	0.04 s	22,336 B		
4	207,909	0.23 s	0.01 s	1 ms	2.14 s	0.07 s	21,696 B		
5	259,189	0.30 s	0.02 s	1 ms	2.48 s	0.08 s	22,528 B		
6	310,469	0.32 s	0.02 s	1 ms	2.96 s	0.08 s	22,272 B	260 B	128 B
7	361,749	0.34 s	0.02 s	1 ms	3.35 s	0.10 s	22,208 B		
8	413,029	0.39 s	0.02 s	1 ms	3.75 s	0.12 s	22,272 B		
9	464,309	0.42 s	0.02 s	1 ms	4.16 s	0.11 s	22,464 B		
10	515,589	0.46 s	0.02 s	1 ms	4.60 s	0.12 s	22,208 B		

TABLE 5. BATCHING q INDEPENDENT MATRIX-MULTIPLICATION CLAIMS $\{\mathbf{A}_i \mathbf{B}_i = \mathbf{C}_i\}_{i \in [q]}$ AT FIXED $k = 2^7$.

B. Thus, on this heterogeneous AI workload, LAMP still reduces prover-side cost, while the sampled opening and linking layers become the main contributors to verification time and proof size.

System	Constraints	Prove	Verify	Proof Size
LAMP	33,427,908	230.82 s	6.52 s	3,342,724 B
Freivalds	76,807,671	408.69 s	0.49 s	196 B

TABLE 6. GPT-2 AT SEQUENCE LENGTH 2^{10} .

8. Conclusion

We proposed LAMP, a protocol for verifying matrix multiplication by moving expensive vector-matrix products outside the SNARK circuit and checking them through sampled local tests over encoded data. The prover commits to the encoded matrices and Freivalds intermediate vectors, while the SNARK circuit proves only the sampled fold relations. Merkle openings and CP-Link proofs ensure that the values used inside the CP-SNARK are consistent with the externally committed data fixed before the sampling challenges are known.

Our implementation confirms the intended tradeoff. For a fixed number t of queries, the main arithmetic relation has $\mathcal{O}(tk + E_{\text{code}})$ constraints rather than the $\mathcal{O}(k^2)$ constraints of a Freivalds baseline. Thus, when t is fixed and the code-consistency checks are linear, the constraint count of LAMP grows linearly with the matrix dimension k . In the matrix benchmark at $k = 4,096$, LAMP reduces the constraint count by $30.3\times$ and proving time by $8.43\times$, while

keeping verification around 0.06 s. The batching experiment further shows that opening overhead can be amortized when multiple matrix-multiplication claims share the same leaf layout.

LAMP also applies to verifiable AI workloads. On a GPT-2 layer, where the computation contains rectangular matrix multiplications of different sizes, the gains are smaller because fewer products can share the same committed leaf layout. Even in this heterogeneous setting, LAMP achieves a $2.30\times$ constraint reduction and a $1.77\times$ proving-time improvement over the Freivalds baseline.

The main remaining costs are sampled-opening proof size and pairing-based link verification. As future work, we plan to reduce opening and linking overhead, improve sharing across consecutive matrix products of different sizes, and exploit fixed model weights in verifiable AI scenarios for preprocessing and batching. These optimizations may reduce proof size and verification time on more complex AI workloads while preserving the linear constraint growth of LAMP.

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Appendix A. SNARK Security Definitions

Let $\mathcal{R} = \{\mathcal{R}_\lambda\}_{\lambda \in \mathbb{N}}$ be a family of NP relations. A SNARK for $\mathcal{R}$ is a tuple of algorithms $\Pi = (\text{Setup}, \text{Prove}, \text{Verify})$, as defined in Section 3.4. We recall the standard security properties used in this work.

Completeness. Π satisfies completeness if, for every security parameter λ and every $(x, w) \in \mathcal{R}_\lambda$,

$$\Pr \left[\begin{array}{l} \Pi.\text{Verify}(\text{pp}, x, \pi) = 1 \\ \text{pp} \leftarrow \Pi.\text{Setup}(1^\lambda, \mathcal{R}_\lambda), \quad \pi \leftarrow \Pi.\text{Prove}(\text{pp}, x, w) \end{array} \right] \geq 1 - \text{negl}(\lambda).$$

If the probability is 1, we say that Π has perfect completeness.

Knowledge soundness. Π satisfies knowledge soundness if, for every PPT prover $\mathcal{A}$, there exists an PPT extractor $\text{Ext}^{\mathcal{A}}$ such that

$$\Pr \left[\begin{array}{l} \Pi.\text{Verify}(\text{pp}, x, \pi) = 1 \\ \wedge (x, w) \notin \mathcal{R}_\lambda \end{array} : \begin{array}{l} \text{pp} \leftarrow \Pi.\text{Setup}(1^\lambda, \mathcal{R}_\lambda), \\ (x, \pi) \leftarrow \mathcal{A}(\text{pp}), \\ w \leftarrow \text{Ext}^{\mathcal{A}}(\text{pp}, x) \end{array} \right] \leq \text{negl}(\lambda).$$

Equivalently, except with negligible probability, any prover that outputs an accepting proof for x must know a witness w such that $(x, w) \in \mathcal{R}_\lambda$.

Zero knowledge. Π satisfies zero knowledge if there exists a PPT simulator Sim such that, for every PPT distinguisher $\mathcal{D}$, the following two experiments are indistinguishable.

$$\text{Real}_{\Pi, \mathcal{D}}(\lambda) : \begin{cases} \text{pp} \leftarrow \Pi.\text{Setup}(1^\lambda, \mathcal{R}_\lambda), \\ \pi \leftarrow \Pi.\text{Prove}(\text{pp}, x, w), \\ b \leftarrow \mathcal{D}(\text{pp}, x, \pi), \\ \text{return } b. \end{cases}$$

$$\text{Ideal}_{\Pi, \mathcal{D}}(\lambda) : \begin{cases} \text{pp} \leftarrow \Pi.\text{Setup}(1^\lambda, \mathcal{R}_\lambda), \\ \pi^* \leftarrow \text{Sim}(\text{pp}, x), \\ b \leftarrow \mathcal{D}(\text{pp}, x, \pi^*), \\ \text{return } b. \end{cases}$$

For all $(x, w) \in \mathcal{R}_\lambda$,

$$|\Pr[\text{Real}_{\Pi, \mathcal{D}}(\lambda) = 1] - \Pr[\text{Ideal}_{\Pi, \mathcal{D}}(\lambda) = 1]| \leq \text{negl}(\lambda).$$

A SNARK satisfying zero knowledge is called a zk-SNARK.

Appendix B. CP-Link Security Definitions

Let $\mathcal{R}_{\text{link}} = \{\mathcal{R}_{\text{link}, \lambda}\}_{\lambda \in \mathbb{N}}$ be the blockwise linking relation defined in Section 3.5. For a SNARK-side key ck_S , an external key ck_E , a SNARK-side commitment $\widetilde{\text{cm}}$, external commitments $\text{cm}_0, \dots, \text{cm}_{q-1}$, sampled blocks $(\mathbf{m}_i)_{i=0}^{q-1}$, and opening randomness $\widetilde{o}, o_0, \dots, o_{q-1}$, define

$$\begin{aligned} x_{\text{link}} &= ((\text{ck}_S, \widetilde{\text{cm}}), (\text{ck}_E, \text{cm}_0, \dots, \text{cm}_{q-1})), \\ w_{\text{link}} &= ((\mathbf{m}_i)_{i=0}^{q-1}, \widetilde{o}, (o_i)_{i=0}^{q-1}). \end{aligned}$$

The relation $\mathcal{R}_{\text{link}, \lambda}$ contains exactly the pairs $(x_{\text{link}}, w_{\text{link}})$ satisfying

$$\begin{aligned} \widetilde{\text{cm}} &= \text{Com}(\text{ck}_S, \mathbf{m}_0 \| \dots \| \mathbf{m}_{q-1}; \widetilde{o}), \\ \text{cm}_i &= \text{Com}(\text{ck}_E, \mathbf{m}_i; o_i) \quad \text{for all } i = 0, \dots, q-1. \end{aligned}$$

A CP-Link proof system for $\mathcal{R}_{\text{link}}$ is a tuple of algorithms $\Pi_{\text{link}} = (\text{Setup}, \text{Prove}, \text{Verify})$, as defined in Section 3.5. The corresponding linking language is

$$\mathcal{L}_{\text{link}, \lambda} = \{x_{\text{link}} : \exists w_{\text{link}} \text{ such that } (x_{\text{link}}, w_{\text{link}}) \in \mathcal{R}_{\text{link}, \lambda}\}.$$

Thus a statement is in $\mathcal{L}_{\text{link}, \lambda}$ exactly when the SNARK-side commitment opens to the ordered concatenation of sampled blocks and each external commitment opens to the corresponding block. The main security property for our protocol is link soundness. If the CP-Link backend is implemented as a NIZK or QA-NIZK, as in our implementation, it may additionally satisfy witness zero knowledge [10], [38].

Link completeness. Π_{link} satisfies completeness if, for every security parameter λ and every $(x_{\text{link}}, w_{\text{link}}) \in \mathcal{R}_{\text{link}, \lambda}$,

$$\Pr \left[\begin{array}{l} \Pi_{\text{link}}.\text{Verify}(\text{pp}_{\text{link}}, x_{\text{link}}, \pi_{\text{link}}) = 1 \\ \text{pp}_{\text{link}} \leftarrow \Pi_{\text{link}}.\text{Setup}(1^\lambda, \mathcal{R}_{\text{link}, \lambda}), \\ \pi_{\text{link}} \leftarrow \Pi_{\text{link}}.\text{Prove}(\text{pp}_{\text{link}}, x_{\text{link}}, w_{\text{link}}) \end{array} \right] \geq 1 - \text{negl}(\lambda).$$

If the probability is 1, we say that Π_{link} has perfect completeness.

Link soundness. Π_{link} satisfies link soundness with error $\epsilon_{\text{link}}(\lambda)$ if, for every PPT adversary $\mathcal{A}$,

$$\Pr \left[\begin{array}{l} \Pi_{\text{link}}.\text{Verify}(\text{pp}_{\text{link}}, x_{\text{link}}^*, \pi_{\text{link}}^*) = 1 \\ \wedge x_{\text{link}}^* \notin \mathcal{L}_{\text{link}, \lambda} \\ \text{pp}_{\text{link}} \leftarrow \Pi_{\text{link}}.\text{Setup}(1^\lambda, \mathcal{R}_{\text{link}, \lambda}), \\ (x_{\text{link}}^*, \pi_{\text{link}}^*) \leftarrow \mathcal{A}(\text{pp}_{\text{link}}) \\ \leq \epsilon_{\text{link}}(\lambda). \end{array} \right]$$

If $\epsilon_{\text{link}}(\lambda) = \text{negl}(\lambda)$, we say that Π_{link} is link-sound. Equivalently, except with probability $\epsilon_{\text{link}}(\lambda)$, an accepting CP-Link proof certifies the existence of blocks $(\mathbf{m}_i)_{i=0}^{q-1}$, an aggregate opening $\tilde{o}$, and external openings $(o_i)_{i=0}^{q-1}$ such that $\tilde{\text{cm}}$ opens to $\mathbf{m}_0 \parallel \dots \parallel \mathbf{m}_{q-1}$ and each cm_i opens to $\mathbf{m}_i$.

Witness zero knowledge. When privacy of the linked blocks and opening randomness is required, we say that Π_{link} satisfies witness zero knowledge if there exists a PPT simulator Sim_{link} such that, for every PPT distinguisher $\mathcal{D}$, the following two experiments are indistinguishable.

$$\begin{aligned} \text{Real}_{\text{link}, \mathcal{D}}(\lambda) : & \begin{cases} \text{pp}_{\text{link}} \leftarrow \Pi_{\text{link}}.\text{Setup}(1^\lambda, \mathcal{R}_{\text{link}, \lambda}), \\ \pi_{\text{link}} \leftarrow \Pi_{\text{link}}.\text{Prove}(\text{pp}_{\text{link}}, x_{\text{link}}, w_{\text{link}}), \\ b \leftarrow \mathcal{D}(\text{pp}_{\text{link}}, x_{\text{link}}, \pi_{\text{link}}), \\ \text{return } b. \end{cases} \\ \text{Ideal}_{\text{link}, \mathcal{D}}(\lambda) : & \begin{cases} \text{pp}_{\text{link}} \leftarrow \Pi_{\text{link}}.\text{Setup}(1^\lambda, \mathcal{R}_{\text{link}, \lambda}), \\ \pi_{\text{link}}^* \leftarrow \text{Sim}_{\text{link}}(\text{pp}_{\text{link}}, x_{\text{link}}), \\ b \leftarrow \mathcal{D}(\text{pp}_{\text{link}}, x_{\text{link}}, \pi_{\text{link}}^*), \\ \text{return } b. \end{cases} \end{aligned}$$

For all $(x_{\text{link}}, w_{\text{link}}) \in \mathcal{R}_{\text{link}, \lambda}$,

$$|\Pr[\text{Real}_{\text{link}, \mathcal{D}}(\lambda) = 1] - \Pr[\text{Ideal}_{\text{link}, \mathcal{D}}(\lambda) = 1]| \leq \text{negl}(\lambda).$$

This property says that the linking proof reveals no information about the blocks $(\mathbf{m}_i)_{i=0}^{q-1}$ or the openings $\tilde{o}, o_0, \dots, o_{q-1}$, beyond the public fact that the commitments are blockwise linkable.

Appendix C. Detailed Proof of Theorem 6.2

Proof. We prove the public-coin interactive protocol first. The Fiat-Shamir variant is handled at the end.

Perfect completeness. Assume that the honest prover holds matrices $\mathbf{A}, \mathbf{B}, \mathbf{C} \in \mathbb{F}^{k \times k}$ with $\mathbf{AB} = \mathbf{C}$. Let $\mathbf{r} = (1, r, \dots, r^{k-1})$, and define

$$\mathbf{x} = \mathbf{rA}, \quad \mathbf{y} = \mathbf{xB}, \quad \mathbf{z} = \mathbf{rC}.$$

Then $\mathbf{y} = \mathbf{z}$. The honest prover commits to the packed encoded matrix blocks $\{\mathbf{m}_{\text{ABC}, j}\}_{j \in [n]}$ under rt_{ABC} , commits to the packed encoded vector blocks $\{\mathbf{m}_{\text{XYZ}, j}\}_{j \in [n]}$ under rt_{XYZ} , and supplies the sampled openings and CP-Link proofs required by Section 5.4.

For every queried position $j \in I$, code linearity gives

$$\text{Fold}(\mathbf{r}, \hat{\mathbf{A}}[*, j]) = \hat{\mathbf{x}}[j],$$

and similarly

$$\text{Fold}(\mathbf{x}, \hat{\mathbf{B}}[*, j]) = \hat{\mathbf{y}}[j], \quad \text{Fold}(\mathbf{r}, \hat{\mathbf{C}}[*, j]) = \hat{\mathbf{z}}[j].$$

Since $\mathbf{y} = \mathbf{z}$, the equality check $\hat{\mathbf{y}}[j] = \hat{\mathbf{z}}[j]$ also holds. All SNARK, Merkle, and CP-Link checks are honestly generated, so the verifier accepts.

Bad-event decomposition. Let Acc be the event that the verifier accepts a false committed matrix multiplication statement. We decompose the failure probability into the following events:

- $\text{Bad}_{\text{SNARK}}$: the CP-SNARK accepts although the sampled relation of Section 5.3 is false for the committed sampled witnesses;
- Bad_{open} : a Merkle opening or Pedersen leaf opening is accepted in a way that violates position binding or commitment binding;
- Bad_{link} : a CP-Link proof accepts although the SNARK-side sampled blocks are not the corresponding blocks opened from the Pedersen leaves under rt_{ABC} and rt_{XYZ} .
- Bad_{code} : the probabilistic code-consistency checks accept encoded views that are not encodings of their corresponding intermediate vectors.

By the assumed soundness of the selected backends,

$$\begin{aligned} \Pr[\text{Bad}_{\text{SNARK}}] &\leq \epsilon_{\text{cp}}(\lambda), \\ \Pr[\text{Bad}_{\text{open}}] &\leq \epsilon_{\text{open}}(\lambda), \\ \Pr[\text{Bad}_{\text{link}}] &\leq \epsilon_{\text{link}}(\lambda), \\ \Pr[\text{Bad}_{\text{code}}] &\leq \epsilon_{\text{code}}(\lambda). \end{aligned}$$

Condition on the complement of these events. Then rt_{ABC} fixes a unique sequence of packed matrix-column commitments at all queried positions, rt_{XYZ} fixes a unique sequence of packed vector-symbol commitments at all queried positions, and the sampled values used by the SNARK are exactly the messages opened from those commitments. In particular, the adversary cannot choose different sampled blocks after seeing I without breaking the opening or linking layer.

Freivalds randomization. The matrix root rt_{ABC} is fixed before r is derived. In the committed statement, rt_{ABC} binds to row-wise encodings of matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}$. Suppose $\mathbf{AB} \neq \mathbf{C}$, and let $\mathbf{D} = \mathbf{AB} - \mathbf{C} \neq 0$. Some column $\mathbf{d}$ of $\mathbf{D}$ is nonzero. The event $\mathbf{rD} = 0$ implies

$$P(r) := \langle (1, r, \dots, r^{k-1}), \mathbf{d} \rangle = \sum_{i=0}^{k-1} d_i r^i = 0.$$

The polynomial P is nonzero and has degree at most $k-1$. Since r is sampled after rt_{ABC} is fixed, Schwartz-Zippel gives

$$\Pr[\mathbf{rD} = 0] \leq \frac{k-1}{|\mathbb{F}|}.$$

Proximity sampling. Now also condition on $\mathbf{rD} \neq 0$. No vectors $\mathbf{x}, \mathbf{y}, \mathbf{z}$ can satisfy all four global relations

$$\mathbf{x} = \mathbf{rA}, \quad \mathbf{y} = \mathbf{xB}, \quad \mathbf{z} = \mathbf{rC}, \quad \mathbf{y} = \mathbf{z}.$$

Otherwise they would imply $\mathbf{rAB} = \mathbf{rC}$, contradicting $\mathbf{rD} \neq 0$.

Therefore at least one checked encoded relation is false. Consider the case $\mathbf{x} \neq \mathbf{rA}$; the other three cases are identical. The codewords $\text{Enc}(\mathbf{x})$ and $\text{Enc}(\mathbf{rA})$ are distinct, so by relative distance they disagree on at least δn coordinates. For any queried coordinate j , code linearity gives

$$\text{Enc}(\mathbf{rA})[j] = \text{Fold}(\mathbf{r}, \hat{\mathbf{A}}[*, j]).$$

The sampled-opening backend ensures that the SNARK uses the opened $\hat{\mathbf{A}}[*, j]$ block under rt_{ABC} and the opened $\hat{\mathbf{x}}[j]$ symbol under rt_{XYZ} . Thus the fold check rejects whenever j lands in the disagreement set. A uniformly sampled coordinate misses this set with probability at most $1 - \delta$, and t independent samples miss it with probability at most $(1 - \delta)^t$. Union-bounding over the four possible violated relations gives

$$4(1 - \delta)^t.$$

Combining the bounds. Adding the SNARK, opening, linking, code-consistency, Freivalds, and proximity failure events gives

$$\Pr[\text{Acc} \mid \mathbf{AB} \neq \mathbf{C}] \leq \epsilon_{\text{cp}}(\lambda) + \epsilon_{\text{open}}(\lambda) + \epsilon_{\text{link}}(\lambda) + \epsilon_{\text{code}}(\lambda) + \frac{k-1}{|\mathbb{F}|} + 4(1 - \delta)^t.$$

For the measured rows, $t = 128$. When $\delta = 1/2$, the proximity term is $4 \cdot 2^{-128} = 2^{-126}$. Larger t or a larger-distance code can be used when the deployment target requires a smaller proximity term.

Fiat–Shamir variant. The non-interactive protocol derives r , I , and backend batch challenges from domain-separated hashes of the transcript. In the random oracle model, the public-coin proof above applies with the standard Fiat–Shamir loss $\epsilon_{\text{FS}}(q_H)$ for an adversary making q_H random-oracle queries. This proves Theorem 6.2. $\square$

Appendix D. Barycentric Reed–Solomon Consistency Check

This appendix specifies the Reed–Solomon instantiation of EncCheck_C used in Section 5.3. Let

$$D = \{\omega_0, \dots, \omega_{n-1}\} \subset \mathbb{F}$$

be the public evaluation domain. A message $\mathbf{v} = (v_0, \dots, v_{k-1}) \in \mathbb{F}^k$ defines the degree- $< k$ polynomial

$$p_{\mathbf{v}}(X) = \sum_{\ell=0}^{k-1} v_{\ell} X^{\ell}, \quad \text{Enc}(\mathbf{v}) = (p_{\mathbf{v}}(\omega_i))_{i=1}^n.$$

Given a claimed encoded word $\hat{\mathbf{v}} \in \mathbb{F}^n$, let $P_{\hat{\mathbf{v}}}$ be the unique degree- $< n$ polynomial satisfying

$$P_{\hat{\mathbf{v}}}(\omega_i) = \hat{\mathbf{v}}[i] \quad \text{for all } i \in [n].$$

The check samples $\alpha \leftarrow \mathbb{F} \setminus D$ after the claimed word is fixed and compares $P_{\hat{\mathbf{v}}}(\alpha)$ with $p_{\mathbf{v}}(\alpha)$.

For efficient evaluation, define the domain polynomial and barycentric weights

$$L_D(X) = \prod_{i=1}^n (X - \omega_i), \quad \lambda_i = \left(\prod_{\ell \neq i} (\omega_i - \omega_{\ell}) \right)^{-1}.$$

For every $\alpha \notin D$, barycentric interpolation gives

$$P_{\hat{\mathbf{v}}}(\alpha) = L_D(\alpha) \sum_{i=1}^n \lambda_i \frac{\hat{\mathbf{v}}[i]}{\alpha - \omega_i}.$$

Thus the concrete Reed–Solomon consistency check is

$$\begin{aligned} \text{EncCheck}_{\text{RS}}(\hat{\mathbf{v}}, \mathbf{v}; \alpha) &= 1 \iff \sum_{\ell=0}^{k-1} v_{\ell} \alpha^{\ell} \\ &= L_D(\alpha) \sum_{i=1}^n \lambda_i \frac{\hat{\mathbf{v}}[i]}{\alpha - \omega_i}. \end{aligned}$$

The right-hand side is a linear form in the claimed encoded symbols once α is fixed. Equivalently, the verifier can derive the interpolation coefficients

$$\gamma_i(\alpha) = L_D(\alpha) \lambda_i (\alpha - \omega_i)^{-1}$$

and the circuit checks

$$\sum_{\ell=0}^{k-1} v_{\ell} \alpha^{\ell} = \sum_{i=1}^n \gamma_i(\alpha) \hat{\mathbf{v}}[i].$$

This costs $\mathcal{O}(k+n)$ field operations per checked point, after the domain-dependent weights are precomputed.

Lemma D.1 (Barycentric code-consistency soundness). *If $\hat{\mathbf{v}} \neq \text{Enc}(\mathbf{v})$ and $\alpha \leftarrow \mathbb{F} \setminus D$ is sampled after $\hat{\mathbf{v}}$ and $\mathbf{v}$ are fixed, then*

$$\Pr[\text{EncCheck}_{\text{RS}}(\hat{\mathbf{v}}, \mathbf{v}; \alpha) = 1] \leq \frac{n-1}{|\mathbb{F}| - n}.$$

Proof. If $\hat{\mathbf{v}} \neq \text{Enc}(\mathbf{v})$, then the interpolation polynomial $P_{\hat{\mathbf{v}}}$ differs from the degree- $< k$ polynomial $p_{\mathbf{v}}$. Their difference

$$Q(X) = P_{\hat{\mathbf{v}}}(X) - p_{\mathbf{v}}(X)$$

is a nonzero polynomial of degree at most $n-1$. The check accepts exactly when $Q(\alpha) = 0$. Since α is uniform over $\mathbb{F} \setminus D$, at most $n-1$ choices of α can make the check accept, which proves the bound. $\square$

In LAMP, this check is applied independently to $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$ at $\alpha_x, \alpha_y, \alpha_z$. A union bound gives the $3(n-1)/(|\mathbb{F}| - n)$ term used in Definition 6.1.

Appendix E. CP-Link Instantiation

This appendix describes the CP-Link instantiation used in our implementation. We use a pairing-based quasi-adaptive link argument, denoted QALink, for linking one

SNARK-side Pedersen commitment to several external Pedersen commitments. The verifier can also batch multiple QALink verification equations into one multi-pairing check.

Let $e : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$ be a bilinear pairing over prime-order groups of order $|\mathbb{F}|$. We write the group operation in $\mathbb{G}_1$ additively, and let $g_2 \in \mathbb{G}_2$ be a generator.

Statement and witness. Fix a block count q and a block length ℓ . Let $\text{ck}_S = ((G_{i,\nu}^S)_{i \in [q], \nu \in [\ell]}, H^S)$ be the SNARK-side commitment key for a commitment to q concatenated blocks, and let $\text{ck}_E = ((G_\nu^E)_{\nu \in [\ell]}, H^E)$ be the external Pedersen commitment key for one block.

The public statement is $\mathbf{x}_{\text{link}} = ((\text{ck}_S, \widetilde{\text{cm}}), (\text{ck}_E, \text{cm}_0, \dots, \text{cm}_{q-1}))$. The witness is

$$\mathbf{w}_{\text{link}} = (\mathbf{m}_0, \dots, \mathbf{m}_{q-1}, \tilde{o}, o_0, \dots, o_{q-1}),$$

where each block $\mathbf{m}_i \in \mathbb{F}^\ell$. The intended relation is that $\widetilde{\text{cm}}$ commits to the ordered concatenation $\mathbf{m}_0 \| \dots \| \mathbf{m}_{q-1}$ under ck_S , while each cm_i commits to the corresponding block $\mathbf{m}_i$ under ck_E . Equivalently,

$$\begin{aligned} \widetilde{\text{cm}} &= \sum_{i \in [q]} \sum_{\nu \in [\ell]} \mathbf{m}_i[\nu] G_{i,\nu}^S + \tilde{o} H^S, \\ \text{cm}_i &= \sum_{\nu \in [\ell]} \mathbf{m}_i[\nu] G_\nu^E + o_i H^E \quad \text{for every } i \in [q]. \end{aligned}$$

Setup. The setup algorithm takes $(\text{ck}_S, \text{ck}_E, q, \ell)$ as input. It samples $k_0, k_1, \dots, k_q, a \leftarrow \mathbb{F}^*$ and defines

$$P_{i,\nu} := k_0 G_{i,\nu}^S + k_{i+1} G_\nu^E \quad (i \in [q], \nu \in [\ell]).$$

It outputs

$$\begin{aligned} \text{pk}_{\text{link}} &= ((P_{i,\nu})_{i \in [q], \nu \in [\ell]}, k_0 H^S, (k_{i+1} H^E)_{i \in [q]}), \\ \text{vk}_{\text{link}} &= (C_0, C_1, \dots, C_q, A) = (ak_0 g_2, ak_1 g_2, \dots, ak_q g_2, ag_2). \end{aligned}$$

The trapdoors $k_0, \dots, k_q, a$ are discarded after setup.

Proving. Given $\mathbf{x}_{\text{link}}$ and $\mathbf{w}_{\text{link}}$, the prover outputs one group element

$$\pi_{\text{link}} := \sum_{i \in [q]} \sum_{\nu \in [\ell]} \mathbf{m}_i[\nu] P_{i,\nu} + \tilde{o}(k_0 H^S) + \sum_{i \in [q]} o_i(k_{i+1} H^E).$$

By construction and by the homomorphism of Pedersen commitments,

$$\pi_{\text{link}} = k_0 \widetilde{\text{cm}} + \sum_{i \in [q]} k_{i+1} \text{cm}_i.$$

This identity is the algebraic relation checked by the verifier.

Verification. The verifier accepts if and only if

$$e(\widetilde{\text{cm}}, C_0) \cdot \prod_{i \in [q]} e(\text{cm}_i, C_{i+1}) = e(\pi_{\text{link}}, A).$$

Equivalently, the verifier checks

$$e(\widetilde{\text{cm}}, C_0) \cdot \prod_{i \in [q]} e(\text{cm}_i, C_{i+1}) \cdot e(\pi_{\text{link}}, -A) = 1_{\mathbb{G}_T}.$$

Correctness. For an honestly generated proof, we have $\pi_{\text{link}} = k_0 \widetilde{\text{cm}} + \sum_{i \in [q]} k_{i+1} \text{cm}_i$. Therefore, by bilinearity,

$$\begin{aligned} e(\pi_{\text{link}}, A) &= e\left(k_0 \widetilde{\text{cm}} + \sum_{i \in [q]} k_{i+1} \text{cm}_i, ag_2\right) \\ &= e(\widetilde{\text{cm}}, ak_0 g_2) \cdot \prod_{i \in [q]} e(\text{cm}_i, ak_{i+1} g_2) \\ &= e(\widetilde{\text{cm}}, C_0) \cdot \prod_{i \in [q]} e(\text{cm}_i, C_{i+1}). \end{aligned}$$

Thus the verifier accepts.

Soundness. QALink is a quasi-adaptive linear proof for the subspace relation induced by the commitment keys $(\text{ck}_S, \text{ck}_E)$, block count q , and block length ℓ . Under the standard quasi-adaptive linear-algebra assumption for the above pairing CRS, any accepting proof for a statement outside this relation implies a nontrivial linear relation among the hidden setup trapdoors $k_0, \dots, k_q, a$. Thus the argument is sound except with negligible probability in the group order.

Verifier cost. For one link statement with q external commitments, verification consists of $q + 2$ pairing terms: one term for $\widetilde{\text{cm}}$, q terms for $\text{cm}_0, \dots, \text{cm}_{q-1}$, and one term for π_{link} against $-A$. A multi-pairing implementation evaluates these terms with $q + 2$ Miller-loop inputs and one final exponentiation. The proof contains one $\mathbb{G}_1$ element, and the verifying key contains $q + 2$ $\mathbb{G}_2$ elements.

Batching pairing equations. Suppose the verifier must check s independent QALink statements. For the h -th statement, let

$$E_h := e(\widetilde{\text{cm}}^{(h)}, C_0^{(h)}) \cdot \prod_{i \in [q_h]} e(\text{cm}_i^{(h)}, C_{i+1}^{(h)}) \cdot e(\pi_{\text{link}}^{(h)}, -A^{(h)}).$$

The unbatched verifier checks $E_h = 1_{\mathbb{G}_T}$ for every h . The batched verifier first derives nonzero coefficients $\lambda_1, \dots, \lambda_s \in \mathbb{F}^*$ by hashing all public statements, verifying keys, proofs, and the surrounding transcript context. It then checks the single equation

$$\prod_{h=1}^s E_h^{\lambda_h} = 1_{\mathbb{G}_T}.$$

Operationally, this is implemented by scaling the $\mathbb{G}_1$ input of every pairing term in the h -th equation by λ_h , and then running one multi-pairing check over all scaled terms:

$$\begin{aligned} &\prod_{h=1}^s e(\lambda_h \widetilde{\text{cm}}^{(h)}, C_0^{(h)}) \cdot \prod_{h=1}^s \prod_{i \in [q_h]} e(\lambda_h \text{cm}_i^{(h)}, C_{i+1}^{(h)}) \\ &\prod_{h=1}^s e(\lambda_h \pi_{\text{link}}^{(h)}, -A^{(h)}) = 1_{\mathbb{G}_T}. \end{aligned}$$

The batched verifier uses $\sum_{h=1}^s (q_h + 2)$ Miller-loop terms and one final exponentiation. It also performs

$\sum_{h=1}^s (q_h + 2)$ scalar multiplications in $\mathbb{G}_1$ to apply the batching coefficients. If all individual equations are valid, then the batched equation is valid by bilinearity. If at least one individual equation is invalid, write $E_h = g_T^{\delta_h}$ in the prime-order target group. The batched equation accepts only if $\sum_h \lambda_h \delta_h = 0$. Since at least one $\delta_h \neq 0$, a fresh random nonzero coefficient vector satisfies this linear equation with probability at most $1/(|\mathbb{F}| - 1)$. In the non-interactive implementation this probability is interpreted in the random oracle model for the Fiat-Shamir-derived batching coefficients.